



Bachelor project

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Stable Distributions

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Abstract

This project gives an introduction to stable distributions and some of their main properties. It begins with the definition of stable laws, their parametrization, and the basic examples: the normal, Cauchy and Lévy distributions. The project then studies the tail behaviour of non-Gaussian stable laws and uses this to determine when moments of stable random variables exist.

The next part concerns the Generalized Central Limit Theorem and explains why stable distributions appear as limits of normalized sums of independent and identically distributed random variables. This is followed by a section on unimodality, where the aim is to show that stable distributions are unimodal.

Finally, stable distributions are related to infinitely divisible distributions through the Lévy–Khintchine representation.

Introduction

In this project, I study stable distributions as a class of probability distributions. The aim is to derive selected fundamental properties, to analyse their tail behaviour and unimodality, and to explain how stable distributions fit into the broader theory of infinitely divisible distributions.

The project is structured as follows. In the first section, I introduce stable laws and derive the basic properties that are needed later in the project. I have kept this section concise by including only the results that are used subsequently. Several of the proofs are based on Nolan (2020). Since some of these arguments are mainly technical and do not add substantial insight to the main exposition, they are either sketched or deferred to the appendix.

The second section studies the tail behaviour of stable distributions. The main proof in this section follows the method presented in Samorodnitsky and Taqqu (1994). The resulting tail estimates are then used to prove properties of stable distributions that were stated, but not proved, in the first section.

The third section introduces one of the main motivations for studying stable distributions: the Generalized Central Limit Theorem. There are several ways to prove this theorem; in this project I follow the approach used by Nolan (2020).

The fourth section is devoted to the unimodality of stable distributions. This is one of the more technically involved parts of the project. Since the literature on this topic is relatively limited, I primarily follow the paper by Simon (2011).

The final section places stable distributions in the broader framework of infinitely divisible distributions. The main result of this section is the Lévy–Khintchine representation. I give both a proof of the representation and an intuitive explanation of the roles of the drift, Gaussian and jump components. The proof follows the approach in Breiman (1992).

Throughout the project, I use standard results from measure theory, probability theory, real analysis and complex analysis. More specifically, I use convergence theorems such as the bounded convergence theorem and the dominated convergence theorem, as well as basic properties of measures and the Lebesgue integral, as covered in LIM. From SAND 1 and SAND 2, I use standard results on convergence in distribution, density transformations and characteristic functions. In particular, many of the proofs rely on basic properties of characteristic functions. In the section on unimodality, I also use some fundamental tools from complex analysis. Since I have not followed a course in complex analysis, these results are taken as background material from Berg (2016) and are used without proof.

1 Basic properties of stable distributions

1.1 Definition of stable laws

Stable laws form a class of probability distributions characterized by the following definition.

Definition 1.1. A non-degenerate random variable X is called stable if, for any $a, b > 0$, there exist constants $c > 0$ and $d \in \mathbb{R}$ such that

$$aX_1 + bX_2 \stackrel{d}{=} cX + d, \quad (1)$$

where X_1 and X_2 are independent copies of X .

It is often easier to work with the following equivalent formulation of stability.

Definition 1.2. A non-degenerate random variable X is stable if and only if for all $n > 1$, there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$X_1 + \cdots + X_n \stackrel{d}{=} c_n X + d_n \quad (2)$$

where $X_1, \dots, X_n$ are independent copies of X .

We will now prove the equivalence of Definition 1.1 and Definition 1.2.

Proof. We start by proving that (1) implies (2). For clarity, we write the constants in (1) as $c(a, b)$ and $d(a, b)$, so that their dependence on a and b is explicit. Thus, for independent copies X_1, X_2 of X , we have

$$aX_1 + bX_2 \stackrel{d}{=} c(a, b)X + d(a, b), \quad a, b > 0.$$

We prove (2) by induction on n . For $n = 2$, (1) with $a = b = 1$ gives

$$X_1 + X_2 \stackrel{d}{=} c(1, 1)X + d(1, 1).$$

Hence (2) holds for $n = 2$ with $c_2 = c(1, 1)$, $d_2 = d(1, 1)$.

Now assume that (2) holds for some $n - 1 \geq 2$, i.e.

$$X_1 + \cdots + X_{n-1} \stackrel{d}{=} c_{n-1}X + d_{n-1}$$

for some $c_{n-1} > 0$ and $d_{n-1} \in \mathbb{R}$. Let $S_{n-1} = X_1 + \cdots + X_{n-1}$. Since X_n is independent of S_{n-1} , and since $S_{n-1} \stackrel{d}{=} c_{n-1}Y + d_{n-1}$ for a copy Y of X , we may write, in distribution, $S_{n-1} + X_n \stackrel{d}{=} c_{n-1}Y + d_{n-1} + Z$, where Y and Z are independent copies of X . Therefore

$$\begin{aligned} X_1 + \cdots + X_n &= S_{n-1} + X_n \stackrel{d}{=} c_{n-1}Y + d_{n-1} + Z \\ &= (c_{n-1}Y + Z) + d_{n-1}. \end{aligned}$$

Applying (1) once more, now with $a = c_{n-1}$ and $b = 1$, gives

$$c_{n-1}Y + Z \stackrel{d}{=} c(c_{n-1}, 1)X + d(c_{n-1}, 1).$$

Hence

$$X_1 + \cdots + X_n \stackrel{d}{=} c(c_{n-1}, 1)X + (d(c_{n-1}, 1) + d_{n-1}).$$

Thus (2) holds for n if we define $c_n = c(c_{n-1}, 1) > 0$, $d_n = d(c_{n-1}, 1) + d_{n-1}$. By induction, (2) holds for all $n > 1$.

The proof of the converse implication follows the same idea, but requires a more careful handling of the constants. We therefore defer the details to Appendix A.1. $\square$

We now identify the scaling constants c_n appearing in (2). They are necessarily of the form

$$c_n = n^{1/\alpha}, \quad (3)$$

for some $0 < \alpha \leq 2$. We do not give the full proof here, but instead outline the main idea. A complete proof is provided in Appendix A.2

Let X be non-degenerate and stable, so that for every $n \geq 1$, $X_1 + \dots + X_n \stackrel{d}{=} c_n X + d_n$, with $c_n > 0$. The argument may first be reduced to the symmetric case by symmetrisation. In that case the centering constants vanish, and the stability relation becomes $X_1 + \dots + X_n \stackrel{d}{=} c_n X$. Comparing a sum of mn independent copies directly with the same sum grouped into m blocks of length n gives $c_{mn} = c_m c_n$, since X is non-degenerate and the scaling constants are positive.

Hence $c_{r^k} = c_r^k$ for every integer $r \geq 2$. Writing $c_r = r^{1/\alpha_r}$, we obtain $c_{r^k} = (r^k)^{1/\alpha_r}$. Thus, along powers of a fixed base r , the norming constants have a power-law form. The remaining point is to show that the exponent α_r is independent of the chosen base. This is done by controlling the ratios c_n/c_ν whenever n and ν are comparable, for instance when $n < \nu \leq rn$. Such a bound rules out different polynomial growth rates along different subsequences.

Indeed, if powers of two different bases r and ρ had different exponents, then the corresponding values of c_n would eventually grow at incompatible rates, even when the indices are of comparable size. Comparing powers $n = r^j$ and $\nu = \rho^k$ with $n < \nu \leq rn$ therefore forces $\alpha_r = \alpha_\rho$. Hence the exponent is common to all bases. Denoting it by α , we obtain $c_n = n^{1/\alpha}$. Moreover, one necessarily has $0 < \alpha \leq 2$.

The last definition of stable distributions which will be given is useful when parameterizing the stable distributions and working with its characteristic functions.

Definition 1.3. A non-degenerate random variable X is stable if and only if $X \stackrel{d}{=} aZ + b$, where $0 < \alpha \leq 2$, $-1 \leq \beta \leq 1$, $a \neq 0, b \in \mathbb{R}$ and Z is a random variable with characteristic function

$$\mathbb{E} \exp(iuZ) = \begin{cases} \exp(-|u|^\alpha [1 - i\beta \tan(\frac{\pi\alpha}{2}) \operatorname{sign}(u)]) , & \alpha \neq 1, \\ \exp(-|u| [1 + i\beta \frac{2}{\pi} \operatorname{sign}(u) \log |u|]) , & \alpha = 1. \end{cases} \quad (4)$$

where $u \in \mathbb{R}$ and $i = \sqrt{-1}$

Proof of equivalence between Definition 1.3 and Definition 1.1.

We prove only the implication from the characteristic function in (4) to the stability property in (1). Let Z_1, Z_2 be independent copies of Z , and let $A, B > 0$.

First assume $\alpha \neq 1$, and write $T_\alpha := \tan(\pi\alpha/2)$. Then $\varphi_{AZ_1+BZ_2}(u) = \varphi_Z(Au)\varphi_Z(Bu)$. Using the characteristic function in (4), we get

$$\varphi_Z(Au)\varphi_Z(Bu) = \exp(-(A^\alpha + B^\alpha)|u|^\alpha [1 - i\beta T_\alpha \operatorname{sgn}(u)]) .$$

Now set $C := (A^\alpha + B^\alpha)^{1/\alpha}$. Since $(A^\alpha + B^\alpha)|u|^\alpha = |Cu|^\alpha$, it follows that $\varphi_Z(Au)\varphi_Z(Bu) = \varphi_Z(Cu)$. We see that $AZ_1 + BZ_2 \stackrel{d}{=} CZ$.

Now let $X = aZ + b$, where $a \neq 0$. If X_1, X_2 are independent copies of X , then

$$AX_1 + BX_2 = a(AZ_1 + BZ_2) + (A + B)b \stackrel{d}{=} aCZ + (A + B)b.$$

Since $CX = C(aZ + b) = aCZ + Cb$, this can be written as $AX_1 + BX_2 \stackrel{d}{=} CX + (A + B - C)b$.

Thus X satisfies the stability property with $C = (A^\alpha + B^\alpha)^{1/\alpha}$, $D = (A + B - C)b$.

For $\alpha = 1$, the same argument gives an additional translation term because of the $\log |u|$ -term in the characteristic function. In this case $C = A + B$, and one obtains

$$AZ_1 + BZ_2 \stackrel{d}{=} CZ + \frac{2\beta}{\pi} (C \log C - A \log A - B \log B) .$$

Consequently, for $X = aZ + b$,

$$AX_1 + BX_2 \stackrel{d}{=} CX + D,$$

where $C = A + B$ and $D = a \frac{2\beta}{\pi} (C \log C - A \log A - B \log B)$. Hence the characteristic function in (4) implies the stability property in Definition 1.1. $\square$

Note that the proof establishes the relation

$$c^\alpha = a^\alpha + b^\alpha \quad (5)$$

for the scaling constants in Definition 1.1. This relation will be useful later when identifying the value of α for specific stable laws.

1.2 Parametrization of stable distributions

Definition 1.3 shows that a general stable distribution requires four parameters: an index of stability $\alpha \in (0, 2]$, a skewness parameter $\beta \in [-1, 1]$, a scale parameter $\gamma > 0$, and a location parameter $\delta \in \mathbb{R}$. These parameters determine different aspects of the distribution.

The role of the parameters can be understood from the characteristic function. Recall that a shift of a random variable corresponds to multiplication of its characteristic function by a factor $e^{i u \delta}$. Thus the term $i u \delta$ in the exponent represents a translation of the distribution and explains why δ is called the location parameter.

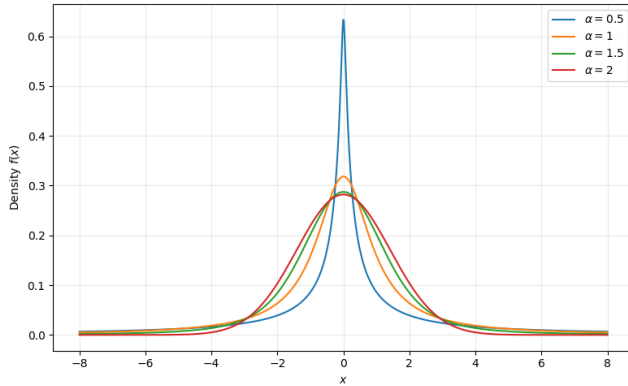
Similarly, scaling a random variable changes the argument of its characteristic function, which is why the scale parameter γ appears together with u through the term $\gamma^\alpha |u|^\alpha$. If $X = \gamma Z + \delta$, then deviations from the location parameter are multiplied by γ . Thus γ controls the scale of the distribution: larger values of γ stretch the distribution around its location parameter, while smaller values compress it.

The parameter α appears as the exponent of $|u|$ in the characteristic function Definition 1.3. It is the same parameter as the one appearing in the scaling relation $c_n = n^{1/\alpha}$ (3). To see why these two appearances of α are related, note that summing n independent copies corresponds to raising the characteristic function in (4) to the n -th power, while multiplying the random variable by c_n corresponds to replacing u by $c_n u$. Comparing the scale terms in the exponent gives

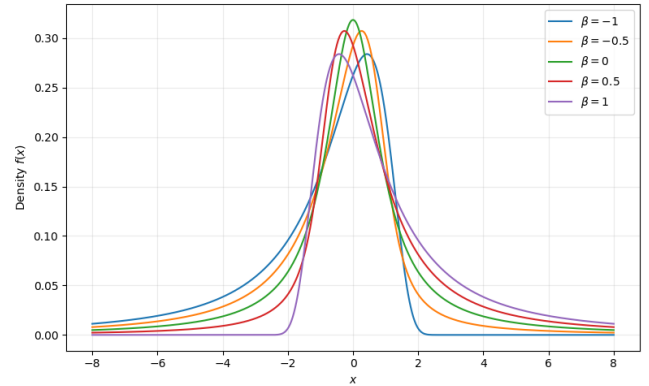
$$n|u|^\alpha = |c_n u|^\alpha = c_n^\alpha |u|^\alpha.$$

Hence $c_n^\alpha = n$, and therefore $c_n = n^{1/\alpha}$. Thus α is both the exponent in the characteristic function and the index governing the growth rate of stable sums. α is therefore a central parameter for the shape of the distribution, especially through its effect on tail behaviour. Smaller values of α correspond to heavier tails; this will be studied in more detail later.

The parameter β appears in the imaginary part, of the exponent and controls the asymmetry of the distribution. When $\beta = 0$, the skewness term disappears and the distribution is symmetric around its location parameter. Positive values of β skew the distribution to the right, while negative values skew it to the left.



(a) Fixed $\beta = 0$, varying α .



(b) Fixed $\alpha = 1$, varying β .

Figure 1: Illustration of the effects of the parameters α and β in the stable family. The parameter β controls skewness and the relative weight of the two tails, while α controls tail-heaviness and concentration around the center.

Stable laws admit several equivalent parameterizations. These mainly differ in how scale and location are normalized, and in how the logarithmic term in the case $\alpha = 1$ (4) is handled. We denote a parametrization by

$$\mathbf{S}(\alpha, \beta, \gamma, \delta; k),$$

where k indicates the chosen parametrization. Throughout this project we most often use the 1-parametrization.

Definition 1.4 (1-parametrization of stable laws). Let $0 < \alpha \leq 2$, $-1 \leq \beta \leq 1$, $\gamma > 0$, and $\delta \in \mathbb{R}$. We say that $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ if

$$X \stackrel{d}{=} \begin{cases} \gamma Z + \delta, & \alpha \neq 1, \\ \gamma Z + \left(\delta + \beta \frac{2}{\pi} \gamma \log \gamma \right), & \alpha = 1, \end{cases}$$

where $Z = Z(\alpha, \beta)$ is the standard stable random variable with characteristic function given in (4). X has characteristic function

$$\mathbb{E} \exp(iuX) = \begin{cases} \exp(-\gamma^\alpha |u|^\alpha [1 - i\beta \tan(\frac{\pi\alpha}{2}) (\text{sign } u)] + i\delta u), & \alpha \neq 1, \\ \exp(-\gamma |u| [1 + i\beta \frac{2}{\pi} (\text{sign } u) \log |u|] + i\delta u), & \alpha = 1. \end{cases} \quad (6)$$

For $\alpha = 1$, the logarithmic term in the characteristic function(4) means that scaling by γ also produces an additional shift term. The correction $\beta \frac{2}{\pi} \gamma \log \gamma$ is included to compensate for this, so that δ remains the location parameter.

The two formulations in Definition 1.3 and Definition 1.4 are equivalent. The proof consists of rewriting the parameters from one notational convention into the other, and is deferred to subsection A.3.

1.3 The normal, Cauchy and Lévy stable laws

Only a few stable laws have densities that can be written in closed form. The classical examples are the normal distribution ($\alpha = 2$), the Cauchy distribution ($\alpha = 1, \beta = 0$), and the Lévy distribution ($\alpha = \frac{1}{2}, \beta = \pm 1$). In this section, we verify the stability property directly for these examples and identify the corresponding stability index α .

To identify the stability index, recall that the proof of the equivalence between Definition 1.3 and Definition 1.1 showed that the scaling constant in Definition 1.1 satisfies $c^\alpha = a^\alpha + b^\alpha$ (5).

Normal distributions are stable since linear combinations of independent normal random variables are again normal. If X_1, X_2 are i.i.d. $N(\mu, \sigma^2)$, then

$$aX_1 + bX_2 \sim N((a+b)\mu, (a^2 + b^2)\sigma^2).$$

This has the same distribution as $cX + d$, where $X \sim N(\mu, \sigma^2)$, by choosing

$$c = \sqrt{a^2 + b^2}, \quad d = (a + b - c)\mu.$$

Therefore by Definition 1.1 the normal distribution is stable. Since $c^2 = a^2 + b^2$, the corresponding stability index is $\alpha = 2$.

We can also identify the location parameter δ and the scale parameter γ in the 1-parametrization from Definition 1.4. When $\alpha = 2$, the skewness term vanishes since $\tan(\pi\alpha/2) = \tan(\pi) = 0$. Hence the characteristic function reduces to

$$\varphi_X(u) = \exp(-\gamma^2 u^2 + i\delta u).$$

Comparing this with the characteristic function of $N(\mu, \sigma^2)$, $\varphi_X(u) = \exp\left(i\mu u - \frac{\sigma^2 u^2}{2}\right)$, gives $\delta = \mu$ and $\gamma^2 = \sigma^2/2$. Thus the normal case in the 1-parametrization is $N(\delta, 2\gamma^2)$. In particular, the parameter β has no effect when $\alpha = 2$, since the term containing β disappears from the characteristic function.

The Cauchy distribution with scale parameter $\gamma > 0$ and location parameter $\delta \in \mathbb{R}$ has density

$$f(x) = \frac{1}{\pi} \frac{\gamma}{\gamma^2 + (x - \delta)^2}, \quad x \in \mathbb{R},$$

and characteristic function $\varphi_X(u) = \exp(i\delta u - \gamma|u|)$.

We verify stability directly. Let X_1, X_2 be independent copies of $X \sim \text{Cauchy}(\gamma, \delta)$, and let $a, b > 0$. Then, by independence,

$$\varphi_{aX_1 + bX_2}(u) = \varphi_X(au) \varphi_X(bu) = e^{ia\delta u - \gamma a|u|} e^{ib\delta u - \gamma b|u|} = e^{i(a+b)\delta u - \gamma(a+b)|u|} = \varphi_{(a+b)X}(u).$$

Hence

$$aX_1 + bX_2 \stackrel{d}{=} (a+b)X.$$

Thus the Cauchy distribution is stable with $c = a + b$ and $d = 0$. Then by (5) we get that $\alpha = 1$. Moreover, after centering at δ ,

$$e^{-i\delta u} \varphi_X(u) = e^{-\gamma|u|},$$

which is real and even. Hence the centred distribution $X - \delta$ is symmetric, so the skewness parameter is $\beta = 0$. The stability property can also be checked directly by using the convolution formula known from SAND 1.

The Lévy distribution $X \sim \text{Levy}(\gamma, \delta)$ has density

$$f(x) = \sqrt{\frac{\gamma}{2\pi}} \frac{1}{(x - \delta)^{3/2}} \exp\left(-\frac{\gamma}{2(x - \delta)}\right), \quad \delta < x < \infty.$$

This is the classical right-skewed Lévy distribution. It corresponds to the stable parameters $\alpha = \frac{1}{2}$ and $\beta = 1$. Its stability can be verified by the same method as in the Cauchy case or using the convolution formula. The sign of β determines the direction of skewness. The density above is supported on (δ, ∞) and corresponds to $\beta = 1$. The case $\beta = -1$ is obtained by replacing the standard variable Z by $-Z$ in (1.4). This reflects the distribution about its location parameter δ and gives the left-skewed Lévy stable law, supported on $(-\infty, \delta)$.

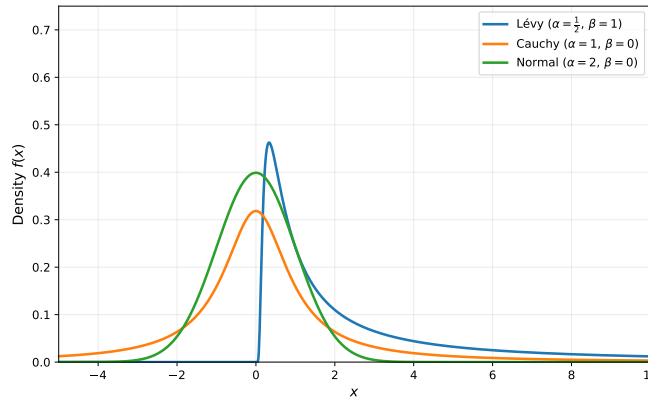


Figure 2: Comparison of the Lévy, Cauchy and normal distributions, illustrating the transition from one-sided heavy tails to symmetric heavy tails and finally light-tailed Gaussian behaviour.

1.4 Scaling and summation

In this section, we present some basic calculation rules for scaling and summing stable laws. These results will be used repeatedly in the proofs throughout the project.

Proposition 1.5. *Let X_1 and X_2 be independent random variables with $X_i \sim \mathbf{S}(\alpha, \beta_i, \gamma_i, \delta_i; 1)$, $i = 1, 2$. Then $X_1 + X_2 \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ with*

$$\gamma = (\gamma_1^\alpha + \gamma_2^\alpha)^{1/\alpha}, \quad \beta = \frac{\beta_1 \gamma_1^\alpha + \beta_2 \gamma_2^\alpha}{\gamma_1^\alpha + \gamma_2^\alpha}, \quad \delta = \delta_1 + \delta_2$$

Proof. The result is shown for the case $\alpha \neq 1$ by considering the characteristic functions and using independence.

$$\begin{aligned} \ln E e^{iu(X_1+X_2)} &= \ln E e^{iuX_1} + \ln E e^{iuX_2} \\ &= -(\gamma_1^\alpha + \gamma_2^\alpha)|u|^\alpha + i|u|^\alpha \text{sign}(u) \tan \frac{\pi\alpha}{2} (\beta_1 \gamma_1^\alpha + \beta_2 \gamma_2^\alpha) + iu(\delta_1 + \delta_2) \\ &= -(\gamma_1^\alpha + \gamma_2^\alpha)|u|^\alpha \left[1 - i \frac{\beta_1 \gamma_1^\alpha + \beta_2 \gamma_2^\alpha}{\gamma_1^\alpha + \gamma_2^\alpha} \text{sign}(u) \tan \frac{\pi\alpha}{2} \right] + iu(\delta_1 + \delta_2). \end{aligned}$$

The case $\alpha = 1$ follows by the same argument.

Proposition 1.6. *Let $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$, and let $a \neq 0$ and $b \in \mathbb{R}$. Then*

$$aX + b \sim \begin{cases} \mathbf{S}(\alpha, \text{sign}(a)\beta, |a|\gamma, a\delta + b; 1), & \alpha \neq 1, \\ \mathbf{S}\left(1, \text{sign}(a)\beta, |a|\gamma, a\delta + b - \frac{2}{\pi}\beta\gamma a \log |a|; 1\right), & \alpha = 1. \end{cases}$$

Proof. The claim follows from the characteristic function representation by identifying the corresponding parameters, as in the proof given in Proposition 1.5 $\square$

Proposition 1.7. *Let $X_1, \dots, X_n$ be independent and identically distributed random variables with $X_j \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$.*

Then

$$X_1 + \dots + X_n \sim \mathbf{S}\left(\alpha, \beta, n^{1/\alpha}\gamma, n\delta; 1\right).$$

Proof. We prove the result by induction on n . For $n = 1$, the statement is trivial. Assume now that the result holds for some $n \in \mathbb{N}$, that is,

$$S_n := X_1 + \dots + X_n \sim \mathbf{S}\left(\alpha, \beta, n^{1/\alpha}\gamma, n\delta; 1\right).$$

Since $X_{n+1} \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ and X_{n+1} is independent of S_n , Proposition 1.5 gives $S_n + X_{n+1} \sim \mathbf{S}(\alpha, \beta_*, \gamma_*, \delta_*; 1)$, where

$$\gamma_* = (n^{1/\alpha}\gamma)^\alpha + \gamma^\alpha = n\gamma^\alpha + \gamma^\alpha = (n+1)\gamma^\alpha.$$

Hence $\gamma_* = (n+1)^{1/\alpha}\gamma$. Moreover,

$$\beta_* = \frac{\beta(n^{1/\alpha}\gamma)^\alpha + \beta\gamma^\alpha}{(n^{1/\alpha}\gamma)^\alpha + \gamma^\alpha} = \frac{\beta n\gamma^\alpha + \beta\gamma^\alpha}{n\gamma^\alpha + \gamma^\alpha} = \beta.$$

Finally, since location parameters add in the $\mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ -parametrization by Proposition 1.5, $\delta_* = n\delta + \delta = (n+1)\delta$. Therefore

$$X_1 + \dots + X_n + X_{n+1} \sim \mathbf{S}\left(\alpha, \beta, (n+1)^{1/\alpha}\gamma, (n+1)\delta; 1\right).$$

By induction, the result holds for every $n \in \mathbb{N}$. $\square$

1.5 Moments of stable distributions

For a stable law with $0 < \alpha < 2$, the absolute moment $\mathbb{E}|X|^p$ is finite for $0 < p < \alpha$, while $\mathbb{E}|X|^p = +\infty$ for $p \geq \alpha$. This statement will be proven in subsection 2.2. Consequently, among stable laws a finite first moment exists precisely when $\alpha > 1$, whereas no stable law with $0 < \alpha < 2$ has finite variance. The remaining case $\alpha = 2$ is the normal distribution, which we know has finite variance. Thus the normal distribution is the only stable law with finite variance.

Proposition 1.8. *When $1 < \alpha \leq 2$, the mean of $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ is $\mathbb{E}[X] = \delta$*

Proof. Let $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ with $\alpha > 1$. Since stable laws with $\alpha > 1$ have a finite first moment, $\mathbb{E}[X]$ exists. Let $X_1, X_2, \dots$ be independent copies of X . Using Proposition 1.7, we obtain

$$X_1 + \dots + X_n \sim \mathbf{S}(\alpha, \beta, n^{1/\alpha}\gamma, n\delta; 1).$$

Therefore by Proposition 1.6

$$\bar{X}_n := \frac{X_1 + \dots + X_n}{n} \sim \mathbf{S}(\alpha, \beta, n^{-1+1/\alpha}\gamma, \delta; 1).$$

Since $\alpha > 1$, we have $n^{-1+1/\alpha} \rightarrow 0$, for $n \rightarrow \infty$. Hence the scale parameter of $\bar{X}_n$ tends to zero, and so $\bar{X}_n \xrightarrow[n \rightarrow \infty]{d} \delta$, where the limit denotes the degenerate distribution concentrated at δ .

On the other hand, by the law of large numbers, $\bar{X}_n \rightarrow \mathbb{E}[X]$ almost surely, and hence also in distribution. Since limits in distribution are unique, it follows that $\mathbb{E}[X] = \delta$. $\square$

2 Tail behaviour

The aim of this section is to study the tail behaviour of stable distributions. The Gaussian case $\alpha = 2$ has light tails and is not the focus here. Instead, we consider non-Gaussian stable laws, that is, the case $0 < \alpha < 2$. These distributions have one or both tails that are asymptotically power laws: one tail in the totally skewed cases $0 < \alpha < 1$ and $\beta = \pm 1$, and both tails otherwise. More precisely, if $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$, then

$$\lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X > \lambda) = C_\alpha \frac{1+\beta}{2} \gamma^\alpha, \quad \lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X < -\lambda) = C_\alpha \frac{1-\beta}{2} \gamma^\alpha.$$

Here C_α is a constant depending only on α . Hence the tails decay as powers of λ , which is why non-Gaussian stable distributions are said to have power-law tails.

2.1 Proof of the asymptotic tail behaviour

We prove the main result in the case of $0 < \alpha < 1$ and $-1 \leq \beta \leq 1$. This case is easier to treat since, for $0 < \alpha < 1$, a stable random variable can be decomposed into two independent one-sided stable random variables with the same α and $\beta = 1$.

Proposition 2.1. *Let $X \sim \mathbf{S}(\alpha, \beta, \gamma, 0; 1)$ with $\alpha < 1$. Then there exist two i.i.d. random variables Y_1 and Y_2 with common distribution $\mathbf{S}(\alpha, 1, \gamma, 0; 1)$ such that*

$$X \stackrel{d}{=} \left(\frac{1+\beta}{2} \right)^{1/\alpha} Y_1 - \left(\frac{1-\beta}{2} \right)^{1/\alpha} Y_2$$

Proof. This can be verified directly by using Proposition 1.5 and Proposition 1.6.

We begin by relating the stable distributions to a compound Poisson random variable. This makes the analysis of stable laws easier. The following proposition will be stated without proof. A proof can be found in Samorodnitsky and Taquq (1994, Prop. 1.2.11).

Proposition 2.2. *Fix $\alpha \in (0, 1)$ and $\xi > 0$. Let N_ξ be a Poisson random variable with mean $\mathbb{E}N_\xi = \xi^{-\alpha}$, and let $Y_{\xi,k}$, $k = 1, 2, \dots$, be i.i.d. positive random variables, independent of N_ξ , with distribution*

$$\mathbb{P}(Y_{\xi,k} > \lambda) = \begin{cases} \xi^\alpha \lambda^{-\alpha}, & \text{if } \lambda > \xi, \\ 1, & \text{if } 0 < \lambda \leq \xi. \end{cases}$$

Then the compound Poisson random variable

$$X_\xi = \sum_{k=1}^{N_\xi} Y_{\xi,k}$$

converges in distribution as $\xi \rightarrow 0$ to a stable random variable $X_0 \sim \mathbf{S}(\alpha, 1, \gamma_0, 0; 1)$, where $\gamma_0^\alpha = \Gamma(1-\alpha) \cos(\frac{\pi\alpha}{2})$. Moreover, the Laplace transform of X_0 is

$$\mathbb{E}[e^{-\rho X_0}] = \exp\{-a_0^\alpha \rho^\alpha\}, \quad \rho \geq 0,$$

where $a_0^\alpha := \Gamma(1-\alpha) = \frac{\gamma_0^\alpha}{\cos(\pi\alpha/2)}$.

To obtain a stable random variable with an arbitrary scale parameter $\gamma > 0$, define $X_\gamma := \frac{\gamma}{\gamma_0} X_0$. Then, by Proposition 1.6, $X_\gamma \sim \mathbf{S}(\alpha, 1, \gamma, 0; 1)$. Its Laplace transform is therefore

$$\mathbb{E}[e^{-\rho X_\gamma}] = \mathbb{E}\left[e^{-\rho(\gamma/\gamma_0)X_0}\right] = \exp\left\{-\Gamma(1-\alpha) \left(\frac{\gamma}{\gamma_0}\right)^\alpha \rho^\alpha\right\} = \exp\left\{-\frac{\gamma^\alpha}{\cos(\pi\alpha/2)} \rho^\alpha\right\}, \quad \rho \geq 0.$$

Equivalently, if $a^\alpha := \frac{\gamma^\alpha}{\cos(\pi\alpha/2)}$, then $\mathbb{E}[e^{-\rho X_\gamma}] = \exp\{-a^\alpha \rho^\alpha\}$, for $\rho \geq 0$.

Observe that for $0 < \alpha < 1$, $\beta = 1$ and $\delta = 0$, the stable random variable X satisfies $X \geq 0$ a.s. We can therefore use a theorem which relates the behaviour of a random variables Laplace transformation near the origin to the

asymptotic behaviour of the tail $P(X > x)$. We state the theorem without proof; see Feller (1971, Theorem XIII.5.4).

In the theorem below and throughout the rest of the project, we write $f(x) \sim g(x)$ for $x \rightarrow a$ for asymptotic equivalence, meaning that

$$\frac{f(x)}{g(x)} \rightarrow 1 \quad \text{for } x \rightarrow a.$$

Theorem 2.3 (Tauberian theorem). *Let $X \geq 0$ be a random variable. Suppose that for some $p > 0$ and $k > 0$,*

$$\int_0^\infty e^{-\lambda x} \mathbb{P}(X > x) dx \sim k\lambda^{-p} \quad \text{as } \lambda \rightarrow 0.$$

Then

$$\mathbb{P}(X > x) \sim \frac{k}{\Gamma(p)} x^{p-1} \quad \text{as } x \rightarrow \infty.$$

This is the classical Tauberian theorem from Feller (1971) applied to the tail function $x \mapsto \mathbb{P}(X > x)$. The required monotonicity condition from the theorem is automatically satisfied for the tail function, since this function is non-increasing.

We are now ready to state and prove the theorem that describes the asymptotic behavior of stable distributions.

Theorem 2.4 (Asymptotic tail behavior). *Let $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$ with $0 < \alpha < 2$. Then*

$$\begin{cases} \lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X > \lambda) = C_\alpha \frac{1+\beta}{2} \gamma^\alpha, \\ \lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X < -\lambda) = C_\alpha \frac{1-\beta}{2} \gamma^\alpha. \end{cases}$$

Where

$$C_\alpha = \begin{cases} \frac{1-\alpha}{\Gamma(2-\alpha) \cos(\pi\alpha/2)}, & \text{if } \alpha \neq 1, \\ \frac{2}{\pi}, & \text{if } \alpha = 1. \end{cases}$$

Proof. We prove the result for $0 < \alpha < 1$ and $-1 \leq \beta \leq 1$.

We first consider the case $0 < \alpha < 1$, $\beta = 1$, $\delta = 0$ and $\gamma > 0$. In order to apply Theorem 2.3, we need the asymptotic behaviour as $\rho \rightarrow 0$ of the Laplace transform of the tail function $\lambda \mapsto \mathbb{P}(X > \lambda)$. Assuming that X has density f , we have

$$\int_0^\infty e^{-\rho\lambda} \mathbb{P}(X > \lambda) d\lambda = \int_0^\infty e^{-\rho\lambda} \left(\int_\lambda^\infty f(x) dx \right) d\lambda.$$

Since the integrand is non-negative, Tonelli's theorem from LIM (Schilling, 2017, Thm 12.8) allows us to interchange the order of integration. Hence

$$\begin{aligned} \int_0^\infty e^{-\rho\lambda} \left(\int_\lambda^\infty f(x) dx \right) d\lambda &= \int_0^\infty f(x) \left(\int_0^x e^{-\rho\lambda} d\lambda \right) dx = \int_0^\infty f(x) \frac{1 - e^{-\rho x}}{\rho} dx \\ &= \frac{1}{\rho} \left(\int_0^\infty f(x) dx - \int_0^\infty e^{-\rho x} f(x) dx \right) = \frac{1 - \mathbb{E}[e^{-\rho X}]}{\rho}. \end{aligned}$$

Using the expression for the Laplace transform from Proposition 2.2, this becomes

$$\int_0^\infty e^{-\rho\lambda} \mathbb{P}(X > \lambda) d\lambda = \frac{1 - \exp(-a^\alpha \rho^\alpha)}{\rho}.$$

To find its behaviour as $\rho \rightarrow 0$, note that $f(x) \sim g(x)$ is equivalent to $f(x) = g(x) + o(g(x))$. By Taylor expansion we see that $1 - e^{-x} \sim x$ as $x \rightarrow 0$. With $x = a^\alpha \rho^\alpha$, this gives

$$1 - \exp(-a^\alpha \rho^\alpha) \sim a^\alpha \rho^\alpha, \quad \rho \rightarrow 0.$$

Therefore

$$\frac{1 - \exp(-a^\alpha \rho^\alpha)}{\rho} \sim a^\alpha \rho^{\alpha-1} = a^\alpha \rho^{-(1-\alpha)}, \quad \rho \rightarrow 0.$$

Thus Theorem 2.3 applies with $p = 1 - \alpha$ and $k = a^\alpha$. We obtain

$$\mathbb{P}(X > \lambda) \sim \frac{a^\alpha}{\Gamma(1-\alpha)} \lambda^{-\alpha} = \frac{\gamma^\alpha}{\Gamma(1-\alpha) \cos(\pi\alpha/2)} \lambda^{-\alpha} = \gamma^\alpha C_\alpha \lambda^{-\alpha}, \quad \lambda \rightarrow \infty.$$

Thus the result is proved for $0 < \alpha < 1$ and $\beta = 1$

We now extend the result to the case $0 < \alpha < 1$ and $-1 < \beta \leq 1$. For simplicity, we first take $\gamma = 1$ and $\delta = 0$. Let $X \sim \mathbf{S}(\alpha, \beta, 1, 0; 1)$. By Proposition 2.1, we may write

$$X = c_1 Y_1 - c_2 Y_2, \quad c_1 = \left(\frac{1+\beta}{2} \right)^{1/\alpha}, \quad c_2 = \left(\frac{1-\beta}{2} \right)^{1/\alpha},$$

where Y_1 and Y_2 are independent random variables with same α as X and $\beta = 1$. Let $\bar{F}(t) := \mathbb{P}(Y_1 > t)$. Then, using independence and the law of total expectation,

$$\begin{aligned} \mathbb{P}(X > \lambda) &= \mathbb{P}(c_1 Y_1 > \lambda + c_2 Y_2) = \mathbb{E} \left[\mathbb{P} \left(Y_1 > \frac{\lambda + c_2 Y_2}{c_1} \mid Y_2 \right) \right] \\ &= \mathbb{E} \left[\bar{F} \left(\frac{\lambda + c_2 Y_2}{c_1} \right) \right]. \end{aligned}$$

From the one-sided case where $0 < \alpha < 1$ and $\beta = 1$, we know that

$$\bar{F}(t) \sim K t^{-\alpha}, \quad t \rightarrow \infty,$$

where $K = C_\alpha$ in the case $\gamma = 1$. Define

$$T_\lambda := \frac{\lambda + c_2 Y_2}{c_1}, \quad Q_\lambda := \frac{\bar{F}(T_\lambda)}{K T_\lambda^{-\alpha}}, \quad H_\lambda := \left(\frac{\lambda}{\lambda + c_2 Y_2} \right)^\alpha Q_\lambda.$$

Then

$$\lambda^\alpha \mathbb{P}(X > \lambda) = \mathbb{E} [K \lambda^\alpha T_\lambda^{-\alpha} Q_\lambda] = K c_1^\alpha \mathbb{E}[H_\lambda].$$

We now analyse the limit as $\lambda \rightarrow \infty$ for a fixed $\omega \in \Omega$. Since Y_2 is finite a.s., we have

$$T_\lambda(\omega) = \frac{\lambda + c_2 Y_2(\omega)}{c_1} \rightarrow \infty, \quad \lambda \rightarrow \infty.$$

Using the one-sided asymptotic $\bar{F}(t) \sim K t^{-\alpha}$ as $t \rightarrow \infty$, we obtain

$$Q_\lambda(\omega) = \frac{\bar{F}(T_\lambda(\omega))}{K T_\lambda(\omega)^{-\alpha}} \rightarrow 1.$$

Moreover,

$$\left(\frac{\lambda}{\lambda + c_2 Y_2(\omega)} \right)^\alpha \rightarrow 1, \quad \lambda \rightarrow \infty.$$

Hence $H_\lambda \rightarrow 1$ a.s.

It remains to justify that we may pass the limit through the expectation, to do this we use the dominated convergence theorem (Schilling, 2017, Thm 12.2) known from LIM.

We start by bounding $H_\lambda = \left(\frac{\lambda}{\lambda + c_2 Y_2} \right)^\alpha Q_\lambda$. Since $Y_2 \geq 0$,

$$0 \leq \left(\frac{\lambda}{\lambda + c_2 Y_2} \right)^\alpha \leq 1.$$

It is therefore enough to bound Q_λ . Define

$$R(t) := \frac{\bar{F}(t)}{Kt^{-\alpha}} = \frac{t^\alpha \bar{F}(t)}{K}, \quad t > 0.$$

Since $\bar{F}(t) \sim Kt^{-\alpha}$, we have $R(t) \rightarrow 1$ as $t \rightarrow \infty$. Hence there exists $t_0 > 0$ such that

$$R(t) \leq 2, \quad t \geq t_0.$$

For $0 < t \leq t_0$, we use $\bar{F}(t) \leq 1$, and obtain

$$R(t) = \frac{t^\alpha \bar{F}(t)}{K} \leq \frac{t_0^\alpha}{K}.$$

Thus, with $M := \max\left\{2, \frac{t_0^\alpha}{K}\right\}$, we have $R(t) \leq M$ for all $t > 0$. Since $Q_\lambda = R(T_\lambda)$ and $T_\lambda > 0$, it follows that $0 \leq H_\lambda \leq M$. The constant M is integrable on the probability space, so the dominated convergence theorem gives

$$\begin{aligned} \lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X > \lambda) &= Kc_1^\alpha \lim_{\lambda \rightarrow \infty} E[H_\lambda] = Kc_1^\alpha E\left[\lim_{\lambda \rightarrow \infty} H_\lambda\right] \\ &= Kc_1^\alpha. \end{aligned}$$

Since $c_1^\alpha = (1 + \beta)/2$, we conclude that

$$\lim_{\lambda \rightarrow \infty} \lambda^\alpha P(X > \lambda) = K \frac{1 + \beta}{2}.$$

Thus, for $\gamma = 1$ and $\delta = 0$,

$$\mathbb{P}(X > \lambda) \sim C_\alpha \frac{1 + \beta}{2} \lambda^{-\alpha}.$$

For general $\gamma > 0$ and $\delta \in \mathbb{R}$, write $X = \gamma Z + \delta$, where $Z \sim \mathbf{S}(\alpha, \beta, 1, 0; 1)$. Then

$$\mathbb{P}(X > \lambda) = \mathbb{P}\left(Z > \frac{\lambda - \delta}{\gamma}\right).$$

Since $(\lambda - \delta)^\alpha \sim \lambda^\alpha$ as $\lambda \rightarrow \infty$, it follows that $\mathbb{P}(X > \lambda) \sim C_\alpha \frac{1 + \beta}{2} \gamma^\alpha \lambda^{-\alpha}$. $\square$

Remarks: To prove the asymptotic tail behavior for $\lambda \rightarrow -\infty$, it suffices to see that $P(X < -x) = P(-X > x)$ and use the reflection property: if $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta)$ then $-X \sim \mathbf{S}(\alpha, -\beta, \gamma, -\delta)$.

2.2 Moment existence via tail behaviour

We can now prove as claimed in subsection 1.5 that for a stable law X with $0 < \alpha < 2$, the absolute moment $\mathbb{E}|X|^p$ is finite for $0 < p < \alpha$, and $\mathbb{E}|X|^p = \infty$ for $p \geq \alpha$.

First we need the formula for absolute moments. Let $y \geq 0$ and $p > 0$. Then

$$y^p = \int_0^y p\lambda^{p-1} d\lambda = \int_0^\infty p\lambda^{p-1} \mathbf{1}_{\{\lambda < y\}} d\lambda. \quad (7)$$

Letting $y = |X|$ and taking expectations gives

$$\mathbb{E}[|X|^p] = \mathbb{E}\left[\int_0^\infty p\lambda^{p-1} \mathbf{1}_{\{\lambda < |X|\}} d\lambda\right].$$

Since the integrand is non-negative, we may apply Tonelli's theorem from LIM and obtain

$$\mathbb{E}[|X|^p] = p \int_0^\infty \lambda^{p-1} \mathbb{P}(|X| > \lambda) d\lambda.$$

We now apply this formula to a stable law X . Splitting the integral at some $A > 0$, we get

$$\mathbb{E}[|X|^p] = p \int_0^A \lambda^{p-1} \mathbb{P}(|X| > \lambda) d\lambda + p \int_A^\infty \lambda^{p-1} \mathbb{P}(|X| > \lambda) d\lambda.$$

The first term is finite, since $\mathbb{P}(|X| > \lambda) \leq 1$ implies $p \int_0^A \lambda^{p-1} \mathbb{P}(|X| > \lambda) d\lambda \leq p \int_0^A \lambda^{p-1} d\lambda = A^p < \infty$. Thus, the convergence of $\mathbb{E}|X|^p$ is determined by the second term.

By Theorem 2.4,

$$\lambda^\alpha \mathbb{P}(|X| > \lambda) = \lambda^\alpha \mathbb{P}(X > \lambda) + \lambda^\alpha \mathbb{P}(X < -\lambda) \rightarrow C_\alpha \frac{1+\beta}{2} \gamma^\alpha + C_\alpha \frac{1-\beta}{2} \gamma^\alpha = C_\alpha \gamma^\alpha$$

as $\lambda \rightarrow \infty$. Let $K = C_\alpha \gamma^\alpha \geq 0$. Then

$$\frac{\mathbb{P}(|X| > \lambda)}{K \lambda^{-\alpha}} \rightarrow 1, \quad \lambda \rightarrow \infty.$$

Hence there exists $A > 0$ such that for all $\lambda \geq A$,

$$\frac{1}{2} K \lambda^{-\alpha} \leq \mathbb{P}(|X| > \lambda) \leq \frac{3}{2} K \lambda^{-\alpha}.$$

Multiplying by λ^{p-1} gives

$$\frac{1}{2} K \lambda^{p-\alpha-1} \leq \lambda^{p-1} \mathbb{P}(|X| > \lambda) \leq \frac{3}{2} K \lambda^{p-\alpha-1}.$$

Therefore

$$\frac{1}{2} K \int_A^\infty \lambda^{p-\alpha-1} d\lambda \leq \int_A^\infty \lambda^{p-1} \mathbb{P}(|X| > \lambda) d\lambda \leq \frac{3}{2} K \int_A^\infty \lambda^{p-\alpha-1} d\lambda.$$

The integral $\int_A^\infty \lambda^{p-\alpha-1} d\lambda$ converges if and only if $p - \alpha - 1 < -1$, i.e. if and only if $p < \alpha$. Hence

$$\mathbb{E}|X|^p < \infty \quad \text{for } 0 < p < \alpha.$$

If $p \geq \alpha$, then $\int_A^\infty \lambda^{p-\alpha-1} d\lambda = \infty$, and the lower bound above implies $\mathbb{E}|X|^p = +\infty$. Thus

$$\mathbb{E}|X|^p < \infty \iff 0 < p < \alpha.$$

We can now prove a useful result which we will use later.

Proposition 2.5. *Let $X \sim \mathbf{S}(\alpha, \beta, \gamma, 0; 1)$. $X \geq 0$ a.s. if and only if $\alpha \in (0, 1)$ and $\beta = 1$*

Proof. We have already shown in Proposition 2.2 that for $\alpha \in (0, 1)$ and $\beta = 1$ $X \geq 0$ a.s. We will now show the converse. Assume that $X \geq 0$. We first show that $\alpha \in (0, 1)$. Since, by definition, $\alpha \in (0, 2]$, it is enough to rule out the cases $\alpha > 1$ and $\alpha = 1$.

Assume first that $\alpha > 1$. By the moment result proved above, applied with $p = 1$, we have $\mathbb{E}|X| < \infty$. Hence X is integrable, and Proposition 1.8 gives $\mathbb{E}[X] = \delta = 0$. Since $X \geq 0$ a.s., we have $|X| = X$, and therefore $\mathbb{E}|X| = \mathbb{E}[X] = 0$. It follows that $X = 0$ a.s., contradicting that X is non-degenerate.

Assume now for contradiction that $\alpha = 1$. Since $X \geq 0$ we have by the moment result proved above that $\mathbb{E}|X|^p < \infty$ for $0 < p < \alpha$. It follows that $\mathbb{E}[X^p] < \infty$ for every $p \in (0, 1)$. Let X_1 and X_2 be independent copies of X . If $\alpha = 1$, stability gives $X_1 + X_2 \stackrel{d}{=} 2X$ by Proposition 1.7, or equivalently

$$\frac{X_1 + X_2}{2} \stackrel{d}{=} X.$$

Choose $p \in (0, 1)$. Since $x \mapsto x^p$ is strictly concave on $[0, \infty)$, Jensen's inequality gives

$$\left(\frac{X_1 + X_2}{2} \right)^p \geq \frac{X_1^p + X_2^p}{2},$$

with equality if and only if $X_1 = X_2$. Define

$$D := \left(\frac{X_1 + X_2}{2} \right)^p - \frac{X_1^p + X_2^p}{2}.$$

Then $D \geq 0$ a.s. Moreover, using $(X_1 + X_2)/2 \stackrel{d}{=} X$ and $X_1, X_2 \stackrel{d}{=} X$,

$$\mathbb{E}[D] = \mathbb{E}[X^p] - \frac{\mathbb{E}[X_1^p] + \mathbb{E}[X_2^p]}{2} = \mathbb{E}[X^p] - \mathbb{E}[X^p] = 0.$$

Hence $D = 0$ a.s. Since equality in Jensen's inequality holds only when $X_1 = X_2$, we get $X_1 = X_2$ a.s. But X_1 and X_2 are independent copies of X , so X must be degenerate, contradicting our assumption. Hence $\alpha = 1$ is impossible.

We have now excluded both $\alpha > 1$ and $\alpha = 1$. Therefore, $\alpha \in (0, 1)$. We now show that $\beta = 1$. By Theorem 2.4,

$$\mathbb{P}(X < -x) \sim C_\alpha \gamma^\alpha \frac{1-\beta}{2} x^{-\alpha}, \quad x \rightarrow \infty,$$

where $C_\alpha > 0$. Since $X \geq 0$ a.s., we have $\mathbb{P}(X < -x) = 0$ for every $x > 0$. Hence the left tail constant must be zero, so $C_\alpha \gamma^\alpha \frac{1-\beta}{2} = 0$. Since X is non-degenerate, $\gamma > 0$, and since $C_\alpha > 0$, it follows that $1 - \beta = 0$. Therefore $\beta = 1$. $\square$

3 Generalized central limit theorem

In this section, we study a central property of stable distributions: any non-degenerate limit of normalized sums of i.i.d. random variables must be stable. We will prove the Generalized central limit theorem (GCLT) and then look at some of the consequences of this theorem.

3.1 Proof of the generalized central limit theorem

To prove the Generalized Central Limit Theorem, the following result is needed.

Theorem 3.1 (Convergence of Types theorem). *Let X_n be a sequence of random variables and $(a_n)_{n \geq 1} > 0$, $(b_n)_{n \geq 1}$ be a sequence of real constants. Assume*

$$X_n \xrightarrow{d} X \quad \text{and} \quad a_n X_n + b_n \xrightarrow{d} Y$$

where neither X nor Y is degenerate. Then $a_n \rightarrow a > 0$ and $b_n \rightarrow b$ and $aX + b \stackrel{d}{=} Y$

This theorem is useful because it shows that different choices of centering and scaling cannot produce essentially different nondegenerate limits. Any two such limits must be of the same type, differing only by an affine transformation.

Remark 3.2. In the following proof, we will use the property: if $X_n \xrightarrow{d} X$, then the characteristic functions φ_n of X_n converge uniformly on bounded intervals to φ_X . A proof can be found in Breiman (1992, Prop. 8.31).

We will use this repeatedly to control terms of the form $\varphi_n(a_n u)$, where both the characteristic function and the argument depend on n . In particular, if $a_n \rightarrow a$ and $u \in \mathbb{R}$ is fixed, then

$$\varphi_n(a_n u) \rightarrow \varphi_X(a u).$$

To see this, note that

$$|\varphi_n(a_n u) - \varphi_X(a u)| \leq |\varphi_n(a_n u) - \varphi_X(a_n u)| + |\varphi_X(a_n u) - \varphi_X(a u)| \rightarrow 0$$

Since $a_n \rightarrow a$, the sequence $(a_n u)$ is bounded, so the first term tends to 0 by the uniform convergence of φ_n to φ_X on bounded intervals. The second term tends to 0 by continuity of φ_X .

Proof of Theorem 3.1. Since $X_n \xrightarrow{d} X$ and $a_n X_n + b_n \xrightarrow{d} Y$, we have

$$\varphi_n(u) \rightarrow \varphi_X(u), \quad \varphi_{a_n X_n + b_n}(u) \rightarrow \varphi_Y(u), \quad \varphi_{a_n X_n + b_n}(u) = e^{i u b_n} \varphi_n(a_n u).$$

Note that $|\varphi_{a_n X_n + b_n}(u)| = |e^{i u b_n} \varphi_n(a_n u)| = |\varphi_n(a_n u)|$.

We start by showing that the sequence $(a_n)_{n \geq 1}$ is bounded. Assume for contradiction that along some subsequence $a_{n_k} \rightarrow \infty$. Because X is nondegenerate, there exists u_0 such that $|\varphi_X(u_0)| < 1$. Define $u_k = u_0/a_{n_k}$. Since $X_{n_k} \xrightarrow{d} X$, we have $\varphi_{n_k}(u_0) \rightarrow \varphi_X(u_0)$, hence

$$|\varphi_{a_{n_k} X_{n_k} + b_{n_k}}(u_k)| = |\varphi_{n_k}(a_{n_k} u_k)| = |\varphi_{n_k}(u_0)| \rightarrow |\varphi_X(u_0)| < 1.$$

On the other hand, since $u_k \rightarrow 0$, (u_k) is bounded and Remark 3.2 gives

$$|\varphi_{a_{n_k} X_{n_k} + b_{n_k}}(u_k)| \rightarrow |\varphi_Y(0)| = 1$$

This is a contradiction, so (a_n) is bounded.

We now show that (a_n) converges. Assume for contradiction that (a_n) does not converge. Since (a_n) is bounded, it follows from the Bolzano-Weierstrass theorem that there exist two subsequences (a_{n_k}) and (a_{m_k}) converging to limits $a \geq 0$ and $a' \geq 0$, respectively, with $a \neq a'$.

We first exclude the possibility that both limits are strictly positive. Suppose $a, a' > 0$. By the remark

$$\varphi_{n_k}(a_{n_k} u) \rightarrow \varphi_X(au), \quad \varphi_{m_k}(a_{m_k} u) \rightarrow \varphi_X(a'u).$$

Also, $e^{iub_n} \varphi_n(a_n u) \rightarrow \varphi_Y(u)$, so along both subsequences

$$e^{iub_{n_k}} \varphi_{n_k}(a_{n_k} u) \rightarrow \varphi_Y(u), \quad e^{iub_{m_k}} \varphi_{m_k}(a_{m_k} u) \rightarrow \varphi_Y(u).$$

Since $|e^{iub_n}| = 1$, it follows that

$$|\varphi_{n_k}(a_{n_k} u)| \rightarrow |\varphi_Y(u)|, \quad |\varphi_{m_k}(a_{m_k} u)| \rightarrow |\varphi_Y(u)|.$$

Together with the subsequential limits, this implies

$$|\varphi_X(au)| = |\varphi_Y(u)| = |\varphi_X(a'u)| \quad \text{for all } u.$$

Assume without loss of generality that $a < a'$ and set $r := a/a' \in (0, 1)$. Writing $v = a'u$, we get from the equation above

$$|\varphi_X(rv)| = |\varphi_X(v)| \quad \text{for all } v.$$

Iterating,

$$|\varphi_X(r^N v)| = |\varphi_X(v)| \quad \text{for all } N \in \mathbb{N}.$$

Since $r^N v \rightarrow 0$, continuity of φ_X yields

$$|\varphi_X(v)| = \lim_{N \rightarrow \infty} |\varphi_X(r^N v)| = |\varphi_X(0)| = 1 \quad \text{for all } v,$$

which is impossible since X is non-degenerate. Hence there cannot be two distinct positive subsequential limits.

Next we show that 0 cannot be a subsequential limit. Suppose that $a_{n_k} \rightarrow 0$ for some subsequence. Fix $u \in \mathbb{R}$. Then $a_{n_k} u \rightarrow 0$, and since $(a_{n_k} u)$ lies in a bounded interval, we may again use uniform convergence on bounded intervals:

$$\varphi_{n_k}(a_{n_k} u) \rightarrow \varphi_X(0) = 1.$$

Therefore

$$|\varphi_Y(u)| = \lim_{k \rightarrow \infty} |e^{iub_{n_k}} \varphi_{n_k}(a_{n_k} u)| = \lim_{k \rightarrow \infty} |\varphi_{n_k}(a_{n_k} u)| = |\varphi_X(0)| = 1 \quad \text{for all } u.$$

This contradicts the fact that Y is non-constant. Thus 0 is not a subsequential limit.

We have shown that (a_n) is bounded, that it cannot have two distinct positive subsequential limits, and that 0 cannot be a subsequential limit. Hence $a_n \rightarrow a$ for some $a > 0$.

We now show that $b_n \rightarrow b$. Since $a_n \rightarrow a$, the remark gives $\varphi_n(a_n u) \rightarrow \varphi_X(au)$ for each fixed u . As $\varphi_X(0) = 1$ and φ_X is continuous, there exists $\delta > 0$ such that $\varphi_X(au) \neq 0$ for all $|u| < \delta$. Hence, for $|u| < \delta$,

$$e^{iub_n} = \frac{e^{iub_n} \varphi_n(a_n u)}{\varphi_n(a_n u)} \rightarrow \frac{\varphi_Y(u)}{\varphi_X(au)}.$$

Thus e^{iub_n} converges for every u in a neighborhood of 0.

Now let b and b' be two subsequential limits of (b_n) . Passing to the limit along the corresponding subsequences gives

$$e^{iub} = e^{iub'} \quad \text{for all } |u| < \delta.$$

Equivalently, $e^{iu(b-b')} = 1$ for all $|u| < \delta$. This is only possible if $b - b' = 0$, and hence $b = b'$. We conclude that (b_n) has at most one subsequential limit.

It remains to show that (b_n) is bounded. Assume otherwise. Then there exists a subsequence (b_{n_k}) with $|b_{n_k}| \rightarrow \infty$. But for any fixed $u \neq 0$, ub_{n_k} becomes unbounded, so the points $e^{iub_{n_k}}$ keep moving around the unit circle and cannot converge on an entire neighborhood of 0. This contradicts the assumed convergence of e^{iub_n} for all $|u| < \delta$. Hence (b_n) must be bounded.

Finally, $e^{iub_n}\varphi_n(a_n u) \rightarrow e^{iub}\varphi_X(au)$, while also $e^{iub_n}\varphi_n(a_n u) \rightarrow \varphi_Y(u)$. Therefore

$$\varphi_Y(u) = e^{iub}\varphi_X(au) = \varphi_{aX+b}(u),$$

and hence $Y \stackrel{d}{=} aX + b$. □

We now turn to the Generalized Central Limit Theorem, which is one of the main motivations for studying stable distributions.

Theorem 3.3 (Generalized Central Limit Theorem). *A non-degenerate random variable Z is stable for some $0 < \alpha \leq 2$ if and only if there is an i.i.d. sequence of random variables X_n and constants $a_n > 0, b_n \in \mathbb{R}$ with*

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z$$

Proof. We first prove that if a normalized sum of i.i.d. random variables converges to a nondegenerate limit, then that limit must be stable.

Let X_n be an i.i.d. sequence of random variables. We set $S_n = X_1 + X_2 + \cdots + X_n$. By assumption there exist constants $a_n > 0$ and b_n such that

$$a_n(X_1 + X_2 + \cdots + X_n) - b_n = a_n S_n - b_n \xrightarrow{d} Z \quad (8)$$

We want to show that Z fulfills Definition 1.2. Fix any integer $n > 1$. We want to split S_n into smaller sums. For $m = 1, 2, 3, \dots$, define $Y_m = a_{nm}S_{nm} - b_{nm}$. Because Y_m is a subsequence of $a_n S_n - b_n$ we get from (8) that $Y_m \xrightarrow{d} Z$ as $m \rightarrow \infty$. Next, S_{nm} is split into n summands $S_{nm} = T_1(m) + T_2(m) + \cdots + T_n(m)$, where each term contains the m variables X_j with indices $j \equiv i \pmod{n}$.

$$T_1(m) = X_1 + X_{n+1} + X_{2n+1} + \cdots + X_{(m-1)n+1},$$

$$\vdots$$

$$T_n(m) = X_n + X_{2n} + X_{3n} + \cdots + X_{mn}.$$

Since the blocks are formed from disjoint sets of i.i.d. variables $T_1(m), \dots, T_n(m)$ are independent. We now rescale Y_m , so that the normalization involves a_m rather than a_{nm} . This allows us to express the sum in terms of the blocks $T_i(m)$ each of which behaves like S_m .

$$\text{Set } A_m = A_m(n) = \frac{a_m}{a_{nm}}, \quad B_m = B_m(n) = \frac{a_m b_{nm}}{a_{nm}} - nb_m.$$

Recall that $Y_m = a_{nm}S_{nm} - b_{nm}$. Then

$$A_m Y_m + B_m = \frac{a_m}{a_{nm}} (a_{nm}S_{nm} - b_{nm}) + \left(\frac{a_m b_{nm}}{a_{nm}} - nb_m \right) = a_m S_{nm} - nb_m.$$

Since $S_{nm} = T_1(m) + T_2(m) + \cdots + T_n(m)$, we obtain

$$\begin{aligned} A_m Y_m + B_m &= a_m (T_1(m) + \cdots + T_n(m)) - nb_m \\ &= (a_m T_1(m) - b_m) + (a_m T_2(m) - b_m) + \cdots + (a_m T_n(m) - b_m). \end{aligned}$$

Because each $T_i(m)$ has the same distribution as S_m ,

$$a_m T_i(m) - b_m \stackrel{d}{=} a_m S_m - b_m \xrightarrow{d} Z_i,$$

where $Z_1, \dots, Z_n$ are i.i.d. copies of Z . Hence

$$A_m Y_m + B_m \xrightarrow{d} Z_1 + Z_2 + \dots + Z_n.$$

On the other hand, from (8) we know that $Y_m \xrightarrow{d} Z$.

We may therefore apply the Convergence of Types theorem. Since

$$Y_m \xrightarrow{d} Z \quad \text{and} \quad A_m Y_m + B_m \xrightarrow{d} Z_1 + \dots + Z_n,$$

there exist constants $C_n > 0$ and D_n such that

$$A_m(n) \rightarrow C_n, \quad B_m(n) \rightarrow D_n, \quad C_n Z + D_n \stackrel{d}{=} Z_1 + \dots + Z_n.$$

Since n was arbitrary, the sum of n i.i.d. copies of Z has the same distribution as an affine transformation of Z . Hence by Definition 1.2 Z is stable.

We have now shown the theorem one way, the other way follows easily. Conversely, suppose that $Z \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$. Let $X_1, X_2, \dots$ be i.i.d. copies of Z .

By the definition of a stable distribution (Definition 1.2), for every $n > 1$ there exist constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that

$$a_n S_n - b_n \stackrel{d}{=} Z, \quad \text{where } S_n = X_1 + \dots + X_n$$

Here, the normalizing constants are given by Proposition 1.7 and Proposition 1.6

$$a_n = n^{-1/\alpha}, \quad b_n = \begin{cases} \delta(n^{1-1/\alpha} - 1), & \alpha \neq 1, \\ \frac{2}{\pi} \beta \gamma \log n, & \alpha = 1. \end{cases}$$

Thus for every n

$$a_n(X_1 + \dots + X_n) - b_n \stackrel{d}{=} Z.$$

This is stronger than (8), since it gives equality in distribution for every n , whereas (8) only asserts convergence in distribution. $\square$

3.2 Domains of attraction

We will now discuss some consequences of the Generalized central limit theorem. For this we need to define domains of attraction.

Definition 3.4. A random variable X is in the domain of attraction of Z if there exist constants $a_n > 0, b_n \in \mathbb{R}$ with

$$a_n(X_1 + \dots + X_n) - b_n \xrightarrow{d} Z$$

where $X_1, X_2, X_3, \dots$ are independent and identically distributed copies of X . $DA(Z)$ is the set of all random variables that are in the domain of attraction of Z .

The Generalized central limit theorem proves that the only possible non-degenerate distributions with a non-empty domain of attraction are stable.

For any $a > 0$ and $b \in \mathbb{R}$,

$$DA(Z) = DA(aZ + b).$$

Indeed, if $a_n S_n - b_n \xrightarrow{d} Z$, then

$$(aa_n)S_n - (ab_n - b) = a(a_n S_n - b_n) + b \xrightarrow{d} aZ + b.$$

The converse follows by applying the same argument to $(x - b)/a$. Hence location and scale do not affect the domain of attraction $DA(Z) = DA(aZ + b)$, so for stable laws it is enough to characterize $DA(Z(\alpha, \beta))$ in terms of α and β .

The next theorem is stated without proof; a proof can be found in Gnedenko and Kolmogorov (1968).

Theorem 3.5. *A random variable X is in the domain of attraction of a Gaussian law if and only if*

$$\frac{t^2 \int_{|x|>t} dP_X(x)}{\int_{|x|<t} x^2 dP_X(x)} \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

We will now show that a non-degenerate random variable with finite variance is in the domain of attraction of a Gaussian law. This is also in accordance with the central limit theorem known from SAND 1.

Assume that X is a random variable and $E[X^2] < \infty$. Set

$$N(t) := t^2 \int_{|x|>t} dP_X = t^2 P(|X| > t), \quad D(t) := \int_{|x|<t} x^2 dP_X = E[X^2 \mathbf{1}_{\{|X|<t\}}]$$

We want to show $\frac{N(t)}{D(t)} \rightarrow 0$.

Since $X^2 \mathbf{1}_{\{|X|>t\}} \rightarrow 0$ a.s. as $t \rightarrow \infty$, and $0 \leq X^2 \mathbf{1}_{\{|X|>t\}} \leq X^2$ with $E[X^2] < \infty$, the dominated convergence theorem yields

$$E[X^2 \mathbf{1}_{\{|X|>t\}}] \rightarrow 0.$$

Moreover, on the set $\{|X| > t\}$ we have $t^2 \leq X^2$, and therefore by monotonicity of the Lebesgue-integral.

$$N(t) = t^2 P(|X| > t) = E[t^2 \mathbf{1}_{\{|X|>t\}}] \leq E[X^2 \mathbf{1}_{\{|X|>t\}}].$$

Hence $N(t) \rightarrow 0$ as $t \rightarrow \infty$. It remains to consider $D(t)$.

$$0 \leq X^2 \mathbf{1}_{\{|X|<t\}} \uparrow X^2 \quad (t \rightarrow \infty)$$

By the monotone convergence theorem, $D(t) = E[X^2 \mathbf{1}_{\{|X|<t\}}] \rightarrow E[X^2]$. Since X is non-degenerate, $D(t) \rightarrow E[X^2] > 0$.

$$\frac{t^2 P(|X| > t)}{E[X^2 \mathbf{1}_{\{|X|<t\}}]} = \frac{N(t)}{D(t)} \rightarrow 0$$

One might wonder whether the domain of attraction of the normal distribution consists only of random variables with finite variance. This is not so: *Pareto*(2, 1) has non-finite variance, yet it lies in the domain of attraction of a Gaussian law. We will return to this example later.

We would like to fully characterize the domains of attraction of stable distributions. To do this, we need the following definition.

Definition 3.6. A function $L(x)$ defined on $(0, \infty)$ is *slowly varying* at infinity if for all $a > 0$

$$\lim_{x \rightarrow \infty} \frac{L(ax)}{L(x)} = 1$$

Theorem 3.7. *Let X be a random variable with distribution function F , and let $0 < \alpha < 2$. Then*

$$X \in DA(Z(\alpha, \beta))$$

if and only if there exist constants

$$c^+ \geq 0, \quad c^- \geq 0, \quad c^+ + c^- > 0,$$

and a slowly varying function L at infinity such that

$$F(-x) \sim c^- x^{-\alpha} L(x), \quad 1 - F(x) \sim c^+ x^{-\alpha} L(x), \quad x \rightarrow \infty.$$

In that case,

$$\beta = \frac{c^+ - c^-}{c^+ + c^-}.$$

A proof can be found in (Breiman, 1992, Thm 9.34). We consider two examples and determine their domains of attraction. First, consider $X \sim \text{Pareto}(\gamma, 1)$, so that $F(x) = 1 - x^{-\gamma}$ and $P(X > x) = x^{-\gamma}$ for $x \geq 1$. Since X is supported on $[1, \infty)$, we have $P(X \leq -x) = 0$ for all sufficiently large x . Thus, for $0 < \gamma < 2$, Theorem 3.7 applies with $L(x) = 1$, $c^+ = 1$ and $c^- = 0$. Hence

$$\beta = \frac{c^+ - c^-}{c^+ + c^-} = 1,$$

and therefore

$$X \in DA(Z(\gamma, 1)), \quad 0 < \gamma < 2.$$

It remains to consider $\gamma \geq 2$. We want to use Theorem 3.5. Since $X \geq 1$,

$$t^2 P(|X| > t) = t^2 P(X > t) = t^{2-\gamma}, \quad \int_{|x| < t} x^2 dP_X(x) = \gamma \int_1^t x^{1-\gamma} dx.$$

For $\gamma = 2$, the ratio in Theorem 3.5 is $\frac{1}{2 \log t} \rightarrow 0$.

For $\gamma > 2$, it is $\frac{t^{2-\gamma}}{\frac{\gamma}{\gamma-2}(1-t^{2-\gamma})} \rightarrow 0$. Hence $X \in DA(Z(2))$ for $\gamma \geq 2$.

Thus

$$X \in \begin{cases} DA(Z(\gamma, 1)), & 0 < \gamma < 2, \\ DA(Z(2)), & \gamma \geq 2. \end{cases}$$

The borderline case $\gamma = 2$ is particularly interesting: here X has non-finite variance, but still belongs to the domain of attraction of a Gaussian law which has finite variance. The non-finite variance follows from the formula for absolute moments (7).

As the next example, we consider $Y = X^2$, where X is a *Cauchy*(0, 1) random variable. We have

$$P(Y > y) = P(X^2 > y) = 1 - P(-\sqrt{y} \leq X \leq \sqrt{y}) = 1 - (F_X(\sqrt{y}) - F_X(-\sqrt{y})).$$

A Cauchy random variable has distribution function

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x).$$

Using that $\arctan(-u) = -\arctan(u)$, we obtain

$$1 - (F_X(\sqrt{y}) - F_X(-\sqrt{y})) = 1 - \left[\left(\frac{1}{2} + \frac{1}{\pi} \arctan(\sqrt{y}) \right) - \left(\frac{1}{2} + \frac{1}{\pi} \arctan(-\sqrt{y}) \right) \right] = 1 - \frac{2}{\pi} \arctan(\sqrt{y}).$$

For $y > 0$, the identity $\arctan(\sqrt{y}) + \arctan\left(\frac{1}{\sqrt{y}}\right) = \frac{\pi}{2}$, gives

$$P(Y > y) = \frac{2}{\pi} \arctan\left(\frac{1}{\sqrt{y}}\right).$$

Since $1/\sqrt{y} \rightarrow 0$ as $y \rightarrow \infty$ and $\arctan(z) \sim z$ as $z \rightarrow 0$, it follows that

$$P(Y > y) \sim \frac{2}{\pi} y^{-1/2}.$$

Thus $1 - F_Y(y) \sim c^+ y^{-\alpha} L(y)$ with $\alpha = \frac{1}{2}$, $c^+ = \frac{2}{\pi}$, and $L(y) \equiv 1$.

Moreover, since $Y = X^2 \geq 0$, we have $F_Y(-y) = 0$ for all $y > 0$, so $c^- = 0$. Hence, by Theorem 3.7,

$$\beta = \frac{c^+ - c^-}{c^+ + c^-} = 1,$$

and therefore $Y \in DA(Z(\frac{1}{2}, 1))$. Thus Y belongs to the domain of attraction of a totally right-skewed stable law with index $\alpha = 1/2$, i.e. a Lévy-stable distribution, see subsection 1.3.

This example is interesting because it starts from a stable random variable, $X \sim \text{Cauchy}(0, 1)$, but the nonlinear transformation $Y = X^2$ changes the tail exponent and places Y in a different domain of attraction, namely

$DA(Z(\frac{1}{2}, 1))$. As we now show, however, this does not mean that Y itself is stable. Assume for contradiction that Y is stable. Since

$$P(Y > y) \sim \frac{2}{\pi} y^{-1/2},$$

Theorem 2.4 implies that the stability index must be $\alpha = \frac{1}{2}$. Moreover, $Y \geq 0$ a.s., so by Proposition 2.5 we must have $\beta = 1$. Hence Y would have to be a Lévy stable law with location parameter 0.

Differentiating the expression $1 - P(Y > y)$ with respect to y gives the resulting density function.

$$f_Y(y) = \frac{1}{\pi \sqrt{y}(1+y)}, \quad y > 0.$$

But a Lévy stable law has density

$$g(y) = Cy^{-3/2}e^{-c/y}, \quad y > 0,$$

for suitable constants $C, c > 0$. These densities have different behaviour near zero: $f_Y(y) \rightarrow \infty$ as $y \rightarrow 0^+$, whereas $g(y) \rightarrow 0$, which can be checked by the substitution $u = 1/y$ and L'Hospital's rule. Thus f_Y cannot be the density of a Lévy stable law, contradicting that Y is stable. Therefore $Y = X^2$ is not stable.

4 Unimodality of stable distributions

A natural next step is to study the densities of stable distributions. Although closed-form expressions are known for only three stable laws, many general qualitative properties of their densities can still be derived.

In this section, we are interested in the number of modes of the density of a stable law. It turns out that all stable laws are unimodal i.e their densities have only one peak.

Definition 4.1. A real random variable X is said to be unimodal if there exists $a \in \mathbb{R}$ such that its distribution function $F(x) = P[X \leq x]$ is convex on $(-\infty, a)$ and concave on $(a, +\infty)$.

The point a is called a mode. It need not to be unique; the uniform distribution is an example.

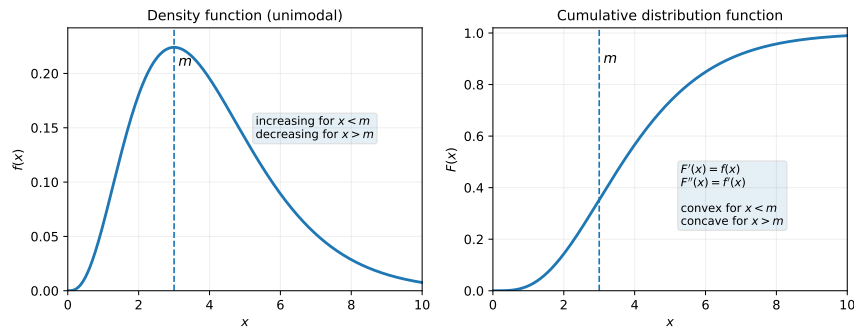


Figure 3: Unimodality illustrated via the density and cumulative distribution functions. The density $f(x)$ increases for $x < a$ and decreases for $x > a$. Since $F'(x) = f(x)$ and $F''(x) = f'(x)$, the cumulative distribution function $F(x)$ is convex for $x < a$ and concave for $x > a$. The point a is the mode and an inflection point of F .

As a simple example, let $X \sim \mathcal{N}(0, 1)$ whose density is $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Then $f'(x) = -xf(x)$. Since $f(x) > 0$ for all x , it follows that $f'(x) > 0$ for $x < 0$ and $f'(x) < 0$ for $x > 0$. Thus X is unimodal with mode $a = 0$.

Theorem 4.2. All stable distributions are unimodal.

We shall not prove this directly for all stable distributions, but instead reduce the argument to the case $0 < \alpha < 1$ and $\beta = 1$. Note that for these parameters $X \geq 0$ a.s. as shown in Proposition 2.5

To prove this case we rewrite the density of stable laws as an integral expression that reveals that the stable law has the same distribution as the product of a transformed independent uniform and a transformed exponential random variable.

4.1 An integral representation of stable density functions

Rather than working directly with the density function, we start by deriving an integral representation for the derivative of the density function.

Lemma 4.3. *Let $0 < \alpha < 1$, $\beta = 1$, and $x > 0$. Let $p(\cdot; \alpha, 1)$ denote the standardized stable density with unit scale and zero location, and write $p'_x(x; \alpha, 1) := \partial_x p(x; \alpha, 1)$. Then*

$$p'_x(x; \alpha, 1) = \frac{1}{\pi} x^{2/(\alpha-1)} \int_0^\pi \exp\left(-x^{\alpha/(\alpha-1)} a(\theta)\right) b(\theta) d\theta,$$

where

$$a(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin\theta}\right)^{1/(1-\alpha)} \frac{\sin((1-\alpha)\theta)}{\sin(\alpha\theta)}$$

and

$$b(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin\theta}\right)^{2/(1-\alpha)} \left(\frac{2\alpha \cos\theta \sin((1-\alpha)\theta)}{(1-\alpha) \sin(\alpha\theta)} - 1\right).$$

It is enough to prove unimodality in the standardised case, since positive rescaling and translation preserve unimodality. Indeed, suppose that X is unimodal then by there exists $m \in \mathbb{R}$ such that its distribution function F_X is convex on $(-\infty, m)$ and concave on (m, ∞) . For $c > 0$ and $d \in \mathbb{R}$, let $Y = cX + d$. Then

$$F_Y(y) = P(Y \leq y) = P\left(X \leq \frac{y-d}{c}\right) = F_X\left(\frac{y-d}{c}\right).$$

Since the map $y \mapsto (y-d)/c$ is increasing and affine, it preserves the convex and concave parts of F_X . Hence F_Y is convex on $(-\infty, cm+d)$ and concave on $(cm+d, \infty)$. Thus Y is unimodal. Hence, for the purpose of proving unimodality, it is enough to consider the standardized case with unit scale and zero location.

Proof of Lemma 4.3. The proof is organized in four steps:

1. We rewrite the integral using Fourier inversion.
2. We differentiate the resulting integral.
3. We deform the contour to obtain a more tractable integral representation.
4. We evaluate the contributions from the different parts of the contour separately.

Step 1: Fourier Inversion

To recover the density from the characteristic function, we use the Fourier inversion formula from Breiman (1992, Thm 8.39). This states that for a random variable with density p and an integrable characteristic function φ , the density is given by, $p(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \varphi(t) dt$.

We start by rewriting this integral. Let

$$I(x) := \int_{-\infty}^{\infty} e^{-itx} \varphi(t) dt = \int_{-\infty}^0 e^{-itx} \varphi(t) dt + \int_0^{\infty} e^{-itx} \varphi(t) dt$$

Thus $p(x) = \frac{1}{2\pi} I(x)$. Set

$$A := \int_0^{\infty} e^{-itx} \varphi(t) dt.$$

Our goal is to show that $\bar{A} = \int_{-\infty}^0 e^{-itx} \varphi(t) dt$. Once this is established, it follows that

$$I = A + \bar{A} = 2\Re(\bar{A}).$$

Note that

$$\overline{e^{-itx}} = \overline{\cos(tx) - i \sin(tx)} = \cos(tx) + i \sin(tx) = e^{itx} \quad (9)$$

and thus

$$\varphi(-t) = E[\exp -itX] = E[\overline{\exp itX}] = \overline{E[\exp itX]} = \overline{\varphi(t)} \quad (10)$$

We have here used the property that $E[\overline{Z}] = \overline{E[Z]}$.

Consider the integral $\int_{-\infty}^0 e^{-itx} \varphi(t) dt$ and substitute $t = -u$:

$$\begin{aligned} \int_{-\infty}^0 e^{-itx} \varphi(t) dt &= \int_0^{\infty} e^{iux} \varphi(-u) du = \int_0^{\infty} \overline{e^{-iux} \varphi(u)} du \\ &= \overline{\int_0^{\infty} e^{-iux} \varphi(u) du} = \overline{A}. \end{aligned}$$

We have here used (9) and (10).

We can now rewrite the inverse Fourier transform as

$$p(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \varphi(t) dt = \frac{1}{\pi} \Re \left(\int_0^{\infty} e^{itx} \varphi(-t) dt \right) \quad (11)$$

To obtain the density through Fourier inversion, we use Zolotarev's B -parameterization of stable laws, as presented in Nolan (2020, Definition 3.6), whose characteristic function is given by

$$E \exp(itX) = \begin{cases} \exp(-\gamma (|t|^\alpha \exp[-i\beta \frac{\pi}{2} K(\alpha)(\text{sign } t)] + i\delta t)), & \alpha \neq 1, \\ \exp(-\gamma (|t| (\frac{\pi}{2} + i\beta(\text{sign } t) \log |t|) + i\delta t)), & \alpha = 1, \end{cases}$$

$$\text{where } K(\alpha) = 1 - |1 - \alpha| = \begin{cases} \alpha, & \alpha < 1, \\ \alpha - 2, & \alpha > 1. \end{cases}$$

We are interested in proving unimodality for the standardized case so $\gamma = 1$ and $\delta = 0$.

To use the fourier inversion formula, we need to verify that the characteristic function is integrable, that is $\int_{\mathbb{R}} |\varphi(u)| du < \infty$.

For $0 < \alpha < 1$, $\beta = 1$, $\gamma = 1$, $\delta = 0$ and $t > 0$, the B -parameterization gives

$$\varphi(t) = \exp \left(-t^\alpha e^{-i\pi\alpha/2} \right).$$

Hence $|\varphi(t)| = \exp \left(-t^\alpha \cos \left(\frac{\pi\alpha}{2} \right) \right)$. The same computation on the negative half-line gives

$$|\varphi(t)| = \exp \left(-|t|^\alpha \cos \left(\frac{\pi\alpha}{2} \right) \right), \quad t \in \mathbb{R}.$$

Since $\cos(\pi\alpha/2) > 0$ for $0 < \alpha < 1$, it follows that $\varphi \in L^1(\mathbb{R})$. Therefore, the Fourier inversion theorem applies, and combined with (11) gives

$$p(x; \alpha, 1) = \frac{1}{\pi} \Re \int_0^{\infty} \exp \left(itx - t^\alpha e^{i\frac{\pi}{2}\alpha} \right) dt.$$

Step 2: Differentiation

To find $p'(x; \alpha, 1)$, we differentiate the above expression for $p(x; \alpha, 1)$ with respect to x . Let

$$u(x, t) := \exp \left(itx - t^\alpha e^{i\pi\alpha/2} \right), \quad (x, t) \in (0, \infty) \times (0, \infty).$$

Then

$$p(x; \alpha, 1) = \frac{1}{\pi} \Re \int_0^{\infty} u(x, t) dt.$$

We want to use the differentiability lemma known from LIM (Schilling, 2017, Thm. 12.5), which lets us differentiate under the integral sign. We therefore need to verify the assumptions of the lemma. First, we need to show that $\int_0^{\infty} |u(x, t)| dt < \infty$. Indeed,

$$|u(x, t)| = \exp \left(\Re \left(itx - t^\alpha e^{i\pi\alpha/2} \right) \right) = \exp \left(-t^\alpha \cos \left(\frac{\pi\alpha}{2} \right) \right).$$

Since $0 < \alpha < 1$, we have $c := \cos\left(\frac{\pi\alpha}{2}\right) > 0$.

Moreover, for any $k > 0$, $\lim_{t \rightarrow \infty} t^k e^{-ct^\alpha} = 0$.

Therefore, there exists $T > 0$ such that for all $t \geq T$,

$$e^{-ct^\alpha} \leq \frac{1}{t^3}.$$

Hence, for $t \geq T$,

$$|u(x, t)| = e^{-ct^\alpha} \leq \frac{1}{t^3},$$

and since $\int_T^\infty t^{-3} dt < \infty$, we obtain $u(x, \cdot) \in L^1(0, \infty)$.

Next, for each fixed $t > 0$, the map $x \mapsto u(x, t)$ is differentiable, with

$$\partial_x u(x, t) = it \exp\left(itx - t^\alpha e^{i\pi\alpha/2}\right), \quad |\partial_x u(x, t)| = te^{-ct^\alpha}.$$

Since $\lim_{t \rightarrow \infty} t^k e^{-ct^\alpha} = 0$ for any $k > 0$, taking $k = 3$ gives some $T > 0$ such that $te^{-ct^\alpha} \leq t^{-2}$ for all $t \geq T$. Hence $te^{-ct^\alpha} \in L^1(0, \infty)$, since it is locally integrable on $(0, T)$ and dominated by t^{-2} on (T, ∞) . Thus $\partial_x u(x, t)$ is dominated by an integrable function independent of x .

By the differentiability lemma,

$$\frac{d}{dx} \left[\frac{1}{\pi} \Re \int_0^\infty u(x, t) dt \right] = \frac{1}{\pi} \Re \int_0^\infty \partial_x u(x, t) dt.$$

Therefore,

$$p'(x; \alpha, 1) = \frac{1}{\pi} \Re \int_0^\infty it \exp\left(itx - t^\alpha e^{i\pi\alpha/2}\right) dt.$$

We now rewrite this integral. Using the substitution $z = xt$, where $x > 0$, we have $t = z/x$, and the change of variables gives

$$\int_0^\infty \exp\left(itx - t^\alpha e^{i\pi\alpha/2}\right) dt = \frac{1}{x} \int_0^\infty \exp\left(iz - z^\alpha x^{-\alpha} e^{i\pi\alpha/2}\right) dz.$$

Here we used

$$itx = i \frac{z}{x} x = iz, \quad t^\alpha e^{i\pi\alpha/2} = \left(\frac{z}{x}\right)^\alpha e^{i\pi\alpha/2} = z^\alpha x^{-\alpha} e^{i\pi\alpha/2}.$$

Thus

$$p'(x; \alpha, 1) = \Re \int_0^\infty \frac{1}{\pi} x^{-2} iz \exp\left(iz - z^\alpha x^{-\alpha} e^{i\pi\alpha/2}\right) dz.$$

Equivalently,

$$p'(x; \alpha, 1) = \Re \int_0^\infty v(z) dz,$$

where

$$v(z) := \frac{1}{\pi} x^{-2} iz \exp\left(iz - z^\alpha x^{-\alpha} e^{i\pi\alpha/2}\right).$$

and

$$z^\alpha = \exp(\alpha \operatorname{Log} z),$$

with $\operatorname{Log} z$ denoting the principal branch of the logarithm on $\mathbb{C} \setminus (-\infty, 0]$. Hence v is well-defined and analytic on $\mathbb{C} \setminus (-\infty, 0]$, since $\operatorname{Log} z$ is analytic there, and therefore $z^\alpha = e^{\alpha \operatorname{Log} z}$ is analytic there as well. As v is obtained by combining analytic functions, it follows that v is analytic on $\mathbb{C} \setminus (-\infty, 0]$.

Step 3: Contour deformation

We want to evaluate

$$\Re \int_0^\infty v(z) dz,$$

but instead of integrating along the positive real axis, we deform the contour to a more convenient path. More precisely, we consider the contour $\Gamma = \Gamma_1 + \Gamma_2$, where Γ_1 is the line segment starting at 0 and ending at $-i\alpha^{1/(1-\alpha)}x^{-\alpha/(1-\alpha)}$, and Γ_2 is the curve given in polar form by

$$z = \rho(\phi)e^{i\phi}, \quad \rho(\phi) = \left(x^{-\alpha} \frac{\sin(\alpha(\frac{\pi}{2} + \phi))}{\sin(\frac{\pi}{2} + \phi)} \right)^{\frac{1}{1-\alpha}}, \quad -\frac{\pi}{2} < \phi < \frac{\pi}{2}.$$

We first check that Γ_1 and Γ_2 fit together continuously. As $\phi \rightarrow -\pi/2$, write

$$u := \phi + \frac{\pi}{2}, \quad u \rightarrow 0^+.$$

Then

$$\frac{\sin(\alpha(\frac{\pi}{2} + \phi))}{\sin(\frac{\pi}{2} + \phi)} = \frac{\sin(\alpha u)}{\sin u} \rightarrow \alpha,$$

since $\sin(\alpha u) \sim \alpha u$ and $\sin u \sim u$ as $u \rightarrow 0$. Hence

$$\rho(\phi) \rightarrow \alpha^{\frac{1}{1-\alpha}} x^{-\frac{\alpha}{1-\alpha}}.$$

Therefore for $\phi \rightarrow -\pi/2$

$$z = \rho(\phi)e^{i\phi} \rightarrow \alpha^{\frac{1}{1-\alpha}} x^{-\frac{\alpha}{1-\alpha}} e^{-i\pi/2} = -i\alpha^{\frac{1}{1-\alpha}} x^{-\frac{\alpha}{1-\alpha}},$$

which is exactly the endpoint of Γ_1 .

Next, as $\phi \rightarrow \pi/2$, we have

$$\theta := \phi + \frac{\pi}{2} \rightarrow \pi.$$

Then $\sin \theta \rightarrow 0$, while $\sin(\alpha\theta) \rightarrow \sin(\alpha\pi) \neq 0$ for $0 < \alpha < 1$. It follows that

$$\frac{\sin(\alpha\theta)}{\sin \theta} \rightarrow \infty,$$

and hence $\rho(\phi) \rightarrow \infty$. Thus Γ_2 extends to infinity. We also note that, apart from the limiting endpoint 0, the contour Γ does not intersect the branch cut $(-\infty, 0]$. Indeed, along Γ_2 the points can be written as $z = re^{i\phi}$, where $r > 0$ and

$$-\frac{\pi}{2} < \phi < \frac{\pi}{2}.$$

Hence $\operatorname{Re} z = r \cos \phi > 0$, so Γ_2 lies in the right half-plane and therefore cannot meet $(-\infty, 0]$. Moreover, Γ_1 lies on the negative imaginary axis, and so it also avoids the branch cut away from the endpoint 0. Therefore v is well-defined along the contour away from the origin.

We now need to justify that the contour deformation is admissible, that is,

$$\int_0^\infty v(z) dz = \int_\Gamma v(z) dz.$$

To justify the contour deformation near the origin, we use a standard cut-off argument. For $\varepsilon > 0$, let Γ_ε denote the original contour with the part inside $|z| < \varepsilon$ removed and replaced by the corresponding circular arc on $|z| = \varepsilon$. On the region enclosed by Γ_ε , the function v is analytic, and therefore Cauchy's integral theorem applies (Berg, 2016, Thm. 3.1).

It remains to let $\varepsilon \downarrow 0$. The added circular arc has length $O(\varepsilon)$, and since $v(z) = O(|z|)$ near the origin, the integral over this arc is $O(\varepsilon^2)$. Hence this contribution tends to zero, so the cut-off contour has the same limiting integral as the original contour.

We also need to justify the cut-off at infinity. Define Γ_n as the part of the contour Γ lying within the circle $|z| = n$, and let $p_n = ne^{i\phi_n}$ be the point where Γ meets the circle $|z| = n$. Define C_n to be the circular arc

$$z(\phi) = ne^{i\phi}, \quad 0 \leq \phi \leq \phi_n,$$

which runs from $z = n$ to p_n .

By Cauchy's integral theorem,

$$\int_0^n v(z) dz + \int_{C_n} v(z) dz - \int_{\Gamma_n} v(z) dz = 0,$$

and hence

$$\int_{\Gamma_n} v(z) dz = \int_0^n v(z) dz + \int_{C_n} v(z) dz.$$

Thus it remains to show that $\int_{C_n} v(z) dz \rightarrow 0$ as $n \rightarrow \infty$. By the triangle inequality,

$$\left| \int_{C_n} v(z) dz \right| \leq \int_{C_n} |v(z)| |dz|.$$

Since C_n lies on the circle $|z| = n$, we parametrize it by $z(\phi) = ne^{i\phi}$, $0 \leq \phi \leq \phi_n$, where ϕ_n is the angle at which the circle meets the deformed contour Γ . Since $z'(\phi) = ine^{i\phi}$, we get

$$\left| \int_{C_n} v(z) dz \right| \leq n \int_0^{\phi_n} |v(ne^{i\phi})| d\phi.$$

Recall that $v(z) := \frac{1}{\pi} x^{-2} iz \exp(iz - z^\alpha x^{-\alpha} e^{i\pi\alpha/2})$. For $z = ne^{i\phi}$, we have $\Re(iz) = -n \sin \phi$, and since the principal branch is used, $z^\alpha = n^\alpha e^{i\alpha\phi}$. Hence

$$\Re(-x^{-\alpha} z^\alpha e^{i\pi\alpha/2}) = -x^{-\alpha} n^\alpha \cos(\alpha(\phi + \pi/2)).$$

Therefore,

$$|v(ne^{i\phi})| = \pi^{-1} x^{-2} n \exp(-n \sin \phi - x^{-\alpha} n^\alpha \cos(\alpha(\phi + \pi/2))).$$

Substituting this into the previous estimate gives

$$\left| \int_{C_n} v(z) dz \right| \leq \pi^{-1} x^{-2} n^2 \int_0^{\phi_n} \exp(-n \sin \phi - x^{-\alpha} n^\alpha \cos(\alpha(\phi + \pi/2))) d\phi.$$

To estimate the integral, we split the interval at a point where the cosine term is still positive. Choose

$$0 < \phi_0 < \min\left\{\frac{\pi}{2}, \frac{\pi(1-\alpha)}{2\alpha}\right\},$$

which is possible since $0 < \alpha < 1$. Then $\alpha(\pi/2 + \phi_0) < \pi/2$, and set

$$c_0 := \cos\left(\alpha\left(\frac{\pi}{2} + \phi_0\right)\right) > 0.$$

Then for $0 \leq \phi \leq \phi_0$, $\cos(\alpha(\frac{\pi}{2} + \phi)) \geq c_0 > 0$. Thus the integrand on $0 \leq \phi \leq \phi_0$ is bounded by a factor $\exp\{-c_0 x^{-\alpha} n^\alpha\}$. We therefore split $[0, \phi_n] = [0, \phi_0] \cup [\phi_0, \phi_n]$ and estimate the two contributions separately.

$$\begin{aligned} \left| \int_{C_n} v(z) dz \right| &\leq \pi^{-1} x^{-2} n^2 \int_0^{\phi_0} \exp[-n \sin \phi - x^{-\alpha} n^\alpha \cos(\alpha(\frac{\pi}{2} + \phi))] d\phi \\ &\quad + \pi^{-1} x^{-2} n^2 \int_{\phi_0}^{\phi_n} \exp\{-n \sin \phi - x^{-\alpha} n^\alpha \cos(\alpha(\frac{\pi}{2} + \phi))\} d\phi. \end{aligned}$$

For the first integral, since $0 \leq \phi \leq \phi_0$ and $\phi_0 < \pi \frac{1-\alpha}{2\alpha}$, we have

$$\alpha\left(\frac{\pi}{2} + \phi\right) \leq \alpha\left(\frac{\pi}{2} + \phi_0\right) < \frac{\pi}{2},$$

and hence

$$\cos(\alpha(\frac{\pi}{2} + \phi)) \geq \cos(\alpha(\frac{\pi}{2} + \phi_0)) > 0.$$

Also $\sin \phi \geq 0$ for $\phi \in (0, \phi_n)$, so

$$-n \sin \phi - x^{-\alpha} n^\alpha \cos(\alpha(\frac{\pi}{2} + \phi)) \leq -x^{-\alpha} n^\alpha \cos(\alpha(\frac{\pi}{2} + \phi_0)).$$

Therefore by $\int_a^b f(\phi)d\phi \leq (b-a) \sup_{\phi \in [a,b]} f(\phi)$, we get

$$\begin{aligned} & \pi^{-1} x^{-2} n^2 \int_0^{\phi_0} \exp\{-n \sin \phi - x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi))\} d\phi \\ & \leq \phi_0 \pi^{-1} x^{-2} n^2 \exp\{-x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi_0))\}. \end{aligned}$$

For the second integral, since $\phi \in [\phi_0, \phi_n]$, we have $\sin \phi \geq \sin \phi_0 > 0$, and we write

$$-n \sin \phi - x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi)) = -n \sin \phi [1 + x^{-\alpha} n^{\alpha-1} \csc \phi \cos(\alpha(\frac{\pi}{2} + \phi))].$$

Where $\csc \phi = 1/\sin \phi$ for $\phi \notin \pi\mathbb{Z}$. Since here $\sin \phi \geq \sin \phi_0 > 0$, this causes no problem on $[\phi_0, \phi_n]$. Now $\sin \phi \geq \sin \phi_0$ implies $\csc \phi \leq \csc \phi_0$, and since $|\cos(\alpha(\frac{\pi}{2} + \phi))| \leq 1$, it follows that

$$|x^{-\alpha} n^{\alpha-1} \csc \phi \cos(\alpha(\frac{\pi}{2} + \phi))| \leq x^{-\alpha} n^{\alpha-1} \csc \phi_0.$$

Because $\alpha < 1$, we have $n^{\alpha-1} \rightarrow 0$, so for all sufficiently large n ,

$$x^{-\alpha} n^{\alpha-1} \csc \phi_0 \leq \frac{1}{2}.$$

For such n , it follows that

$$1 + x^{-\alpha} n^{\alpha-1} \csc \phi \cos(\alpha(\frac{\pi}{2} + \phi)) \geq \frac{1}{2},$$

and therefore

$$-n \sin \phi - x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi)) \leq -\frac{1}{2} n \sin \phi \leq -\frac{1}{2} n \sin \phi_0.$$

Consequently,

$$\begin{aligned} & \pi^{-1} x^{-2} n^2 \int_{\phi_0}^{\phi_n} \exp\{-n \sin \phi - x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi))\} d\phi \\ & \leq \pi^{-1} x^{-2} n^2 \int_{\phi_0}^{\phi_n} \exp\{-n \sin \phi [1 + x^{-\alpha} n^{\alpha-1} \csc \phi \cos(\alpha(\frac{\pi}{2} + \phi))]\} d\phi \\ & \leq \pi^{-1} x^{-2} n^2 \int_{\phi_0}^{\phi_n} e^{-\frac{1}{2} n \sin \phi_0} d\phi \leq \pi^{-1} x^{-2} n^2 \phi_n e^{-\frac{1}{2} n \sin \phi_0}. \end{aligned}$$

In the last inequality, we have used the identity $\int_a^b f(\phi)d\phi \leq (b-a) \sup_{\phi \in [a,b]} f(\phi)$. Combining the two estimates, we conclude that

$$\left| \int_{C_n} v(z) dz \right| \leq \phi_0 \pi^{-1} x^{-2} n \exp\{-x^{-\alpha} n^{\alpha} \cos(\alpha(\frac{\pi}{2} + \phi_0))\} + \pi^{-1} x^{-2} n^2 \phi_n e^{-\frac{1}{2} n \sin \phi_0}.$$

Both terms tend to 0 as $n \rightarrow \infty$, since $\lim_{t \rightarrow \infty} t^k e^{-ct^\alpha} = 0$, for $c > 0$, $\alpha > 0$, $k \geq 0$. Hence

$$\int_{C_n} v(z) dz \rightarrow 0, \quad n \rightarrow \infty,$$

which proves the desired limit. We have now shown that

$$\lim_{n \rightarrow \infty} \int_{\Gamma_n} v(z) dz = \int_0^\infty v(z) dz = \int_{\Gamma_1} v(z) dz + \int_{\Gamma_2} v(z) dz$$

Step 4: Evaluation of the contour contributions

We first show that $\Re \int_{\Gamma_1} v(z) dz = 0$. Recall that $v(z) = \frac{1}{\pi} x^{-2} i z \exp(iz - z^\alpha x^{-\alpha} e^{i\pi\alpha/2})$.

Parametrize Γ_1 by $z(y) = -iy$ for $0 \leq y \leq c$, where $c = \alpha^{1/(1-\alpha)} x^{-\alpha/(1-\alpha)}$. Then

$$\Re \int_{\Gamma_1} v(z) dz = \int_0^c \Re(v(z(y)) z'(y)) dy.$$

where $z'(y) = -i$. It is enough to show that

$$\Re(v(z(y)) z'(y)) = 0 \quad \text{for all } y \in [0, c].$$

Since $z(y) = -iy$, we have $iz(y) = y$, and from $z(y) = ye^{-i\pi/2}$ it follows that

$$z(y)^\alpha = y^\alpha e^{-i\pi\alpha/2}.$$

Thus

$$iz(y) - z(y)^\alpha x^{-\alpha} e^{i\pi\alpha/2} = y - y^\alpha x^{-\alpha} \in \mathbb{R},$$

so the exponential factor in $v(z(y))$ is real. Furthermore,

$$iz(y) z'(y) = -iy$$

is purely imaginary. Hence $v(z(y)) z'(y)$ is purely imaginary on $[0, c]$, and therefore $\Re \int_{\Gamma_1} v(z) dz = 0$.

Next we evaluate the contribution from Γ_2 , namely

$$\Re \int_{\Gamma_2} v(z) dz.$$

Recall that Γ_2 is parametrized by

$$z_{\Gamma_2}(\phi) = \rho(\phi) e^{i\phi}, \quad -\frac{\pi}{2} < \phi < \frac{\pi}{2},$$

where $\rho(\phi)^{1-\alpha} = x^{-\alpha} \frac{\sin(\alpha(\frac{\pi}{2} + \phi))}{\sin(\frac{\pi}{2} + \phi)}$.

The remaining step is to re-parametrize the integral above and simplify the resulting trigonometric expression. The details are deferred to A.4. The resulting integral is

$$\Re \int_{\Gamma_2} v(z) dz = \pi^{-1} x^{2/(\alpha-1)} \int_0^\pi \exp\{-x^{\alpha/(\alpha-1)} a(\theta)\} b(\theta) d\theta, \quad (12)$$

where

$$a(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin\theta} \right)^{1/(1-\alpha)} \frac{\sin((1-\alpha)\theta)}{\sin(\alpha\theta)}$$

and

$$b(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin\theta} \right)^{2/(1-\alpha)} \left(\frac{2\alpha \cos\theta \sin((1-\alpha)\theta)}{(1-\alpha) \sin(\alpha\theta)} - 1 \right).$$

which is the desired representation. $\square$

We have now established an expression for $p'(x; \alpha, 1)$. We next turn to a corresponding representation for $p(x; \alpha, 1)$.

Lemma 4.4. *For a stable random variable with $0 < \alpha < 1$ and $\beta = 1$,*

$$p(x; \alpha, 1) = \frac{\alpha}{(1-\alpha)\pi} x^{\frac{1}{\alpha-1}} \int_0^\pi e^{-x^{\frac{\alpha}{\alpha-1}} a(\varphi)} a(\varphi) d\varphi,$$

where

$$a(\varphi) = \left(\frac{\sin(\alpha\varphi)}{\sin\varphi} \right)^{\frac{1}{1-\alpha}} \frac{\sin((1-\alpha)\varphi)}{\sin(\alpha\varphi)}, \quad 0 < \varphi < \pi.$$

Proof. Define

$$q(x) := \frac{\alpha}{(1-\alpha)\pi} x^{\frac{1}{\alpha-1}} \int_0^\pi e^{-x^{\frac{\alpha}{\alpha-1}} a(\varphi)} a(\varphi) d\varphi.$$

A direct differentiation shows that $q'(x) = p'(x; \alpha, 1)$, where $p'(x; \alpha, 1)$ is of the form established in Lemma 4.3. Hence $q(x) - p(x; \alpha, 1)$ is constant in x . Since both $q(x) \rightarrow 0$ and $p(x; \alpha, 1) \rightarrow 0$ as $x \rightarrow \infty$, the constant is zero and therefore $q(x) = p(x; \alpha, 1)$, which proves the claim. $\square$

4.2 Factorization of the stable density

Now that we have derived the integral representation of the density function for the case $0 < \alpha < 1$ and $\beta = 1$, unimodality may be proved in two different ways. One is a direct analytic approach, based on studying the integral representation, as in Ibragimov and Linnik (1971, Theorem 2.5.3). The other is to use a probabilistic factorization argument. In what follows, we adopt the latter approach and factorize the stable random variable into two independent random variables, following Simon (2011).

Lemma 4.5. *Let L and U be independent random variables where $L \sim \text{Exp}(1)$ with density $\exp(-x)$ for $x > 0$ and $U \sim \text{Unif}(0, \pi)$. Then for $\alpha \in (0, 1)$, $p(x; \alpha, 1)$ is the density of $T_\alpha := \left(\frac{a(U)}{L}\right)^{(1-\alpha)/\alpha}$.*

Here $p(x; \alpha, 1)$ and $a(x)$ are the functions defined in Lemma 4.4

Proof. Let

$$h(x) = \frac{1}{\pi} \int_0^\pi a(\phi) e^{-x a(\phi)} d\phi, \quad x > 0.$$

We first show that h is the density of $X := L/a(U)$, where $L \sim \text{Exp}(1)$ and $U \sim \text{Unif}(0, \pi)$ are independent. Conditioning on $U = \phi$, we have $X|U = \phi = L/a(\phi)$. Since $L \sim \text{Exp}(1)$, it follows that $L/a(\phi)$ is exponentially distributed with rate $a(\phi)$, so

$$f_{X|U=\phi}(x) = a(\phi) e^{-a(\phi)x}, \quad x \geq 0.$$

Therefore, by the law of total probability in density form,

$$f_X(x) = \int_0^\pi f_{X|U=\phi}(x) f_U(\phi) d\phi = \frac{1}{\pi} \int_0^\pi a(\phi) e^{-a(\phi)x} d\phi = h(x),$$

since $f_U(\phi) = 1/\pi$ on $[0, \pi]$.

Now define

$$Y := \left(\frac{a(U)}{L}\right)^{(1-\alpha)/\alpha} = X^{-(1-\alpha)/\alpha}.$$

The map $g(x) = x^{-(1-\alpha)/\alpha}$ is monotone on $(0, \infty)$, with inverse $S(y) := g^{-1}(y) = y^{-\alpha/(1-\alpha)}$. Moreover,

$$|S'(y)| = \frac{\alpha}{1-\alpha} y^{-1/(1-\alpha)}.$$

Hence, by the transformation formula for densities known from SAND 1,

$$f_Y(y) = f_X(S(y)) |S'(y)| = h\left(y^{-\alpha/(1-\alpha)}\right) \frac{\alpha}{1-\alpha} y^{-1/(1-\alpha)} = p(y; \alpha, 1).$$

where the last equality is the representation of $p(y; \alpha, 1)$ from Lemma 4.4. Thus Y has density $p(\cdot; \alpha, 1)$. $\square$

We now rewrite the factorization in a form that is more convenient for later use. Consider T_α from Lemma 4.5.

$$T_\alpha = \left(\frac{a(U)}{L}\right)^{(1-\alpha)/\alpha},$$

Here $U \sim \text{Unif}(0, \pi)$ and $L \sim \text{Exp}(1)$ are independent. Writing $U = \pi V$ with $V \sim \text{Unif}(0, 1)$ allows us to work on the unit interval and absorb the trigonometric terms into a single function b_α .

Now write $U = \pi V$, where $V \sim \text{Unif}(0, 1)$. Then

$$T_\alpha = \left(\frac{a(\pi V)}{L}\right)^{(1-\alpha)/\alpha} = L^{(\alpha-1)/\alpha} a(\pi V)^{(1-\alpha)/\alpha}.$$

Define

$$b_\alpha(v) = \left(\frac{\sin(\pi v)}{\sin(\alpha \pi v)}\right)^\alpha \left(\frac{\sin(\pi v)}{\sin((1-\alpha)\pi v)}\right)^{1-\alpha}, \quad v \in (0, 1).$$

Then

$$a(\pi v) = b_\alpha(v)^{-1/(1-\alpha)},$$

and therefore $a(\pi V)^{(1-\alpha)/\alpha} = b_\alpha(V)^{-1/\alpha}$. Consequently,

$$T_\alpha = L^{(\alpha-1)/\alpha} b_\alpha(V)^{-1/\alpha}. \quad (13)$$

We have now obtained a useful factorization of T_α as a product of two independent random variables.

4.3 Analysis of the resulting factorization

Our strategy is to analyze the two factors separately and then apply a unimodality result for products of random variables. We begin with $b_\alpha(V)^{-1/\alpha}$. Our aim is to show that this random variable has a non-increasing density, since any random variable with a non-increasing density is unimodal. Indeed, if f is non-increasing on its support, then the corresponding distribution function F is concave on its support, since $F'(x) = f(x)$ and hence $F''(x) = f'(x) \leq 0$. It follows that the distribution is unimodal, with mode at the left endpoint of its support.

We will take for granted that b_α is positive, $b'_\alpha(u) < 0$, and concave on $(0, 1)$; a proof may be found in Simon (2012).

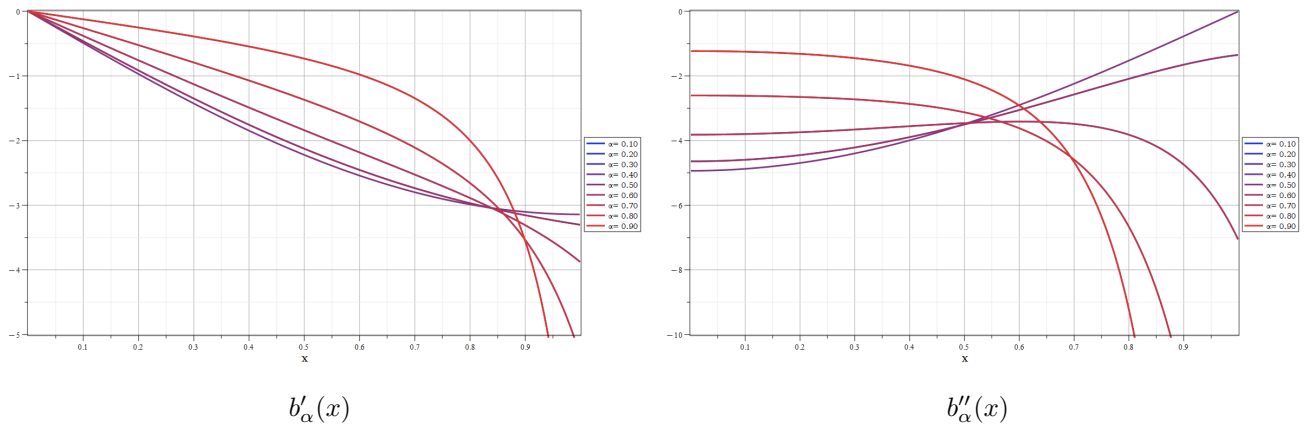


Figure 4: Plots of $b'_\alpha(x)$ and $b''_\alpha(x)$, illustrating concavity.

We now show that $u \mapsto b_\alpha(u)^{-1/\alpha}$ is increasing and convex on $(0, 1)$, given that b_α is positive, $b'_\alpha(u) < 0$ and concave on $(0, 1)$.

Define

$$h(u) := b_\alpha(u)^{-1/\alpha},$$

where $0 < \alpha < 1$. Since $b_\alpha(u) > 0$, this is well-defined.

First, h is increasing. Indeed,

$$h'(u) = -\frac{1}{\alpha} b_\alpha(u)^{-1/\alpha-1} b'_\alpha(u).$$

Since $b_\alpha(u) > 0$, we have $b_\alpha(u)^{-1/\alpha-1} > 0$, and $b'_\alpha(u) < 0$. Therefore

$$h'(u) > 0,$$

so h is strictly increasing.

Next, we show that h is convex. Differentiating once more gives

$$h''(u) = -\frac{1}{\alpha} \left[\left(-\frac{1}{\alpha} - 1 \right) b_\alpha(u)^{-1/\alpha-2} (b'_\alpha(u))^2 + b_\alpha(u)^{-1/\alpha-1} b''_\alpha(u) \right].$$

Rearranging this, we obtain

$$h''(u) = \frac{1}{\alpha} \left(\frac{1}{\alpha} + 1 \right) b_\alpha(u)^{-1/\alpha-2} (b'_\alpha(u))^2 - \frac{1}{\alpha} b_\alpha(u)^{-1/\alpha-1} b''_\alpha(u).$$

Now $b_\alpha(u) > 0$, so both $b_\alpha(u)^{-1/\alpha-2}$ and $b_\alpha(u)^{-1/\alpha-1}$ are positive. Also $(b'_\alpha(u))^2 \geq 0$, and since b_α is concave, $b''_\alpha(u) \leq 0$. Hence both terms on the right-hand side are nonnegative, and therefore $h''(u) \geq 0$. It has now been shown that $u \mapsto b_\alpha(u)^{-1/\alpha}$ is increasing and convex on $(0, 1)$.

We can now use these properties to study the random variable obtained by transforming a uniform variable through h , and show that $h(V)$ has a non-increasing density.

Let

$$Y := h(V) = b_\alpha(V)^{-1/\alpha}, \quad V \sim \text{Unif}(0, 1).$$

Since h is strictly increasing, it is invertible on its image, and for $y \in h((0, 1))$,

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(h(V) \leq y) = \mathbb{P}(V \leq h^{-1}(y)) = h^{-1}(y).$$

Differentiating, we obtain the density of Y :

$$f_Y(y) = (h^{-1})'(y) = \frac{1}{h'(h^{-1}(y))}, \quad y \in h((0, 1)).$$

Now h is convex, so h' is increasing. Since h is also increasing, h^{-1} is increasing as well. Therefore the map

$$y \mapsto h'(h^{-1}(y))$$

is increasing, and hence $y \mapsto \frac{1}{h'(h^{-1}(y))}$ is non-increasing. Thus f_Y is a non-increasing density on its support. A random variable with non-increasing density must be unimodal and therefore the random variable $b_\alpha(V)^{-1/\alpha}$ is unimodal.

We now turn to the factor $L^{(\alpha-1)/\alpha}$ from (13), where $L \sim \text{Exp}(1)$.

We will use the following lemma, which we state without proof; see Theorem 3.7 in Cuculescu and Theodorescu (1998).

Lemma 4.6. *Let X be a positive random variable with density f . If $x \mapsto f(e^x)$ is log-concave on $\mathbb{R}$, then for every unimodal random variable Y , independent of X , the product XY is unimodal. In particular, X itself is unimodal.*

We know that $b_\alpha(V)^{-1/\alpha}$ is unimodal, so by Lemma 4.6 it remains to verify that the density for $L^{(\alpha-1)/\alpha}$

$$x \mapsto f_{L^{(\alpha-1)/\alpha}}(e^x)$$

is log-concave on $\mathbb{R}$. Recall that for a positive function f , log-concavity means that $\log f$ is concave. Since $f_{L^{(\alpha-1)/\alpha}}(e^x) > 0$, it is therefore enough to show that $x \mapsto \log(f_{L^{(\alpha-1)/\alpha}}(e^x))$ is concave.

Set

$$X := L^{(\alpha-1)/\alpha} = L^c, \quad c := \frac{\alpha-1}{\alpha} < 0.$$

Since $L \sim \text{Exp}(1)$, we have

$$f_L(\ell) = e^{-\ell}, \quad \ell > 0.$$

Now $X = L^c$, so the inverse transformation is

$$S(x) := x^{1/c}.$$

By the transformation formula for densities known from SAND 1, we get

$$f_X(e^x) = \left| \frac{\alpha}{\alpha-1} \right| e^{(\alpha/(\alpha-1)-1)x} \exp\left(-e^{\alpha x/(\alpha-1)}\right).$$

Taking logarithms, we obtain

$$\log(f_X(e^x)) = \log\left|\frac{\alpha}{\alpha-1}\right| + \left(\frac{\alpha}{\alpha-1} - 1\right)x - e^{\alpha x/(\alpha-1)}.$$

Differentiating twice yields

$$\frac{d^2}{dx^2} \log(f_X(e^x)) = -\left(\frac{\alpha}{\alpha-1}\right)^2 e^{\alpha x/(\alpha-1)} < 0.$$

Therefore $x \mapsto f_X(e^x)$ is log-concave on $\mathbb{R}$. By Lemma 4.6, it follows that

$$T_\alpha = L^{(\alpha-1)/\alpha} b_\alpha(V)^{-1/\alpha}$$

is unimodal. This concludes the proof that stable laws with $0 < \alpha < 1$ and $\beta = 1$ are unimodal as stated in Theorem 4.2.

5 Infinitely divisible distributions

In this section, we study stable laws within the broader class of infinitely divisible distributions. We show that every stable law is infinitely divisible. We then derive the Lévy–Khintchine representation, which provides a complete characterization of infinitely divisible distributions through their characteristic functions.

5.1 Definition and properties of infinitely divisible distributions

Throughout this chapter, for each $n \in \mathbb{N}$, let

$$S_n = X_1^{(n)} + \cdots + X_n^{(n)} \tag{14}$$

where, for each n , the random variables $X_1^{(n)}, \dots, X_n^{(n)}$ are independent and identically distributed. The notation $X_k^{(n)}$ indicates that the distribution may depend on n .

A natural question is: if $S_n \xrightarrow{D} X$, for $n \rightarrow \infty$, what are the possible distributions of X ?

For normalized sums of the form

$$\frac{X_1 + \cdots + X_n}{A_n} - B_n,$$

the Generalized Central Limit Theorem (Theorem 3.3) states that any non-degenerate limit distribution must be stable. More generally, the possible limit distributions of sums S_n of the form (14) are precisely the infinitely divisible distributions. In this sense, infinitely divisible distributions form the natural broader framework in which stable laws appear as a distinguished subclass.

Definition 5.1. A non-degenerate random variable X is infinitely divisible if for every $n > 1$, there are i.i.d. random variables $X_1^{(n)}, \dots, X_n^{(n)}$ such that

$$X \stackrel{d}{=} X_1^{(n)} + \cdots + X_n^{(n)}$$

This definition does not require that the summands are of the same type as X , whereas the definition for stability does require this; see Definition 1.2. Since stability implies that, for every n , X can be represented as a sum of n i.i.d. random variables after an affine rescaling, every stable law is infinitely divisible.

The Poisson distribution is an example of an infinitely divisible distribution that is not a stable law. Let $X \sim \text{Poisson}(\lambda)$. For any $n \in \mathbb{N}$, let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}\left(\frac{\lambda}{n}\right).$$

Since the sum of independent Poisson random variables is Poisson with parameter equal to the sum of the individual parameters, we get $X_1 + \cdots + X_n \sim \text{Poisson}(\lambda)$. Hence X is infinitely divisible.

However, this construction does not show stability. Stability would require sums of i.i.d. copies of X itself to be reducible to the law of X by an affine rescaling. If $Y_1, \dots, Y_n$ are i.i.d. copies of X , then their sum has distribution $\text{Poisson}(n\lambda)$, which cannot be obtained from $X \sim \text{Poisson}(\lambda)$ by an affine rescaling when $n > 1$. Thus the Poisson distribution is infinitely divisible but not stable.

We aim to show that limits of sums of the form (14) are necessarily infinitely divisible. For this purpose, we use a convergence result whose key assumption is that the family of distributions does not lose mass in the limit. This condition is made precise by the notion of a mass-preserving family.

Definition 5.2. Let $\mathcal{N}$ denote the set of all distribution functions. A subset $\mathcal{L} \subset \mathcal{N}$ is said to be mass-preserving if, for every $\epsilon > 0$, there exists a finite interval I such that

$$F(I^c) < \epsilon \quad \text{for all } F \in \mathcal{L}.$$

With this definition, we can state the convergence result that will be used repeatedly throughout this section.

Proposition 5.3. Let $\mathcal{L} \subset \mathcal{N}$. Then $\mathcal{L}$ is mass-preserving if and only if it is conditionally compact, that is, whenever $F_n \in \mathcal{L}$, there exists a subsequence F_{n_m} and some $F \in \mathcal{N}$ such that

$$F_{n_m} \xrightarrow{D} F.$$

A full proof can be found in Breiman (1992, Prop 8.10)

We can now prove that any limit in distribution of sums of the form (14) is infinitely divisible.

Proposition 5.4. A random variable X has an infinitely divisible distribution if and only if there exists a sequence of sums S_n , defined as in (14), such that

$$S_n \xrightarrow{d} X.$$

Proof. By definition, if X is infinitely divisible, then for each n there exists a sum S_n of the form (14) with distribution equal to that of X .

Conversely, assume that $S_n \xrightarrow{D} X$, for a sequence of sums S_n of the form (14).

We first show that X can be decomposed into the sum of two independent and identically distributed random variables. For $n \geq 1$, write

$$S_{2n} = (X_1^{(2n)} + \cdots + X_n^{(2n)}) + (X_{n+1}^{(2n)} + \cdots + X_{2n}^{(2n)}) = Y_n + Y'_n.$$

Then Y_n and Y'_n are independent and identically distributed.

We now show that the sequence (Y_n) is mass-preserving. Since $S_n \xrightarrow{D} X$, Proposition 5.3 implies that $(S_n)_{n \geq 1}$, and hence also $(S_{2n})_{n \geq 1}$, is mass-preserving. Moreover, $\{Y_n > y, Y'_n > y\} \subseteq \{S_{2n} > 2y\}$, so by independence,

$$(P(Y_n > y))^2 = P(Y_n > y, Y'_n > y) \leq P(S_{2n} > 2y).$$

Similarly,

$$(P(Y_n < -y))^2 \leq P(S_{2n} < -2y).$$

The tails of Y_n are therefore bounded by the tails of the mass-preserving sequence $(S_{2n})_{n \geq 1}$. Hence $(Y_n)_{n \geq 1}$ is mass-preserving.

Applying Proposition 5.3 to the mass-preserving family $(Y_n)_{n \geq 1}$, we may pass to a subsequence (n_k) such that $Y_{n_k} \xrightarrow{d} Y$ for some random variable Y . Since Y'_{n_k} has the same distribution as Y_{n_k} , we also have $Y'_{n_k} \xrightarrow{d} Y'$, where Y' is a copy of Y . Moreover, Y_{n_k} and Y'_{n_k} are independent for each k , and therefore $(Y_{n_k}, Y'_{n_k}) \xrightarrow{d} (Y, Y')$, where Y and Y' are independent. Hence, by the continuous mapping theorem,

$$Y_{n_k} + Y'_{n_k} \xrightarrow{d} Y + Y'.$$

But $S_{2n_k} = Y_{n_k} + Y'_{n_k}$, and since $S_n \xrightarrow{D} X$, it follows in particular that $S_{2n_k} \xrightarrow{D} X$. On the other hand, we have shown that $Y_{n_k} + Y'_{n_k} \xrightarrow{D} Y + Y'$. Therefore $X \stackrel{d}{=} Y + Y'$, where Y and Y' are independent and identically distributed.

The same argument applies for any $m \in \mathbb{N}$. Instead of splitting S_{2n} into two independent blocks, we split S_{mn} into m independent and identically distributed blocks. Repeating the argument above, along a suitable subsequence (n_k) , gives

$$X \stackrel{d}{=} Y_1 + \cdots + Y_m,$$

where $Y_1, \dots, Y_m$ are independent and identically distributed. Since this holds for arbitrary m , X is infinitely divisible by Definition 5.1. $\square$

5.2 The Lévy–Khintchine representation

All infinitely divisible distributions can be characterized by the Lévy–Khintchine representation.

Theorem 5.5 (Lévy–Khintchine representation). *X has an infinitely divisible distribution if and only if its characteristic function $\varphi(u)$ is given by*

$$\log \varphi(u) = i\beta u - \frac{\sigma^2 u^2}{2} + \int \left(e^{iux} - 1 - \frac{iux}{1+x^2} \right) \frac{1+x^2}{x^2} \nu(dx) \quad (15)$$

where ν is a finite measure that assigns zero mass to the origin.

The Lévy–Khintchine representation gives a useful way of interpreting an infinitely divisible distribution through the idea of a random walk. If a random variable X has an infinitely divisible distribution, then for every n it can be represented in distribution as $X \stackrel{d}{=} X_1^{(n)} + \dots + X_n^{(n)}$, where the $X_{i,n}$'s are independent and identically distributed. In this sense, the distribution of X can be viewed as the distribution of the final position of a random walk after many small independent increments.

In the representation above, the three terms correspond to three different types of contributions to the distribution. The term $i\beta u$ is the drift term. It represents a deterministic movement, or the average direction in which the random walk moves when the individual increments have a systematic tendency.

The term $-\frac{\sigma^2 u^2}{2}$ is the Gaussian part. It represents the contribution coming from the accumulation of many small independent fluctuations. This is analogous to the classical central limit theorem: when many small finite-variance fluctuations are added together, and no single jump dominates, their aggregate effect is Gaussian. Thus this term represents the continuous finite-variance fluctuation part of the distribution.

The integral term represents the jump part of the distribution. In a random walk, some increments may be rare but large. In the limit, these rare increments appear as jumps, and the measure $\frac{1+x^2}{x^2} \nu(dx)$ describes the frequency of jumps of different sizes. This part is especially important for distributions with heavy tails. For example, stable laws with $\alpha < 2$ do not have finite variance, and their random behaviour cannot be explained purely by a Gaussian term. Instead, their tail behaviour is captured by the jump part of the Lévy–Khintchine representation. In contrast, the case $\alpha = 2$ corresponds to the normal distribution, where the Gaussian part is the relevant contribution.

Thus the Lévy–Khintchine formula can be read as saying that every infinitely divisible distribution is built from three basic mechanisms: a deterministic drift, a Gaussian finite-variance fluctuation part, and a jump part which allows for rare large movements and heavy-tailed behaviour.

5.3 Proof of the Lévy–Khintchine representation

We now want to prove Theorem 5.5.

We first need the following proposition which is stated without proof, a proof can be found in Breiman (1992, Prop 9.10).

Proposition 5.6. *If $S_n \xrightarrow{D} X$, then $X_1^{(n)} \xrightarrow{D} 0$.*

Proof of Theorem 5.5. We first show that any characteristic function whose logarithm has the form of (15) corresponds to an infinitely divisible distribution.

Assume that $\varphi(u)$ is the characteristic function of a random variable X and that $\log \varphi(u)$ is of the form given in (15). For $n \in \mathbb{N}$, define φ_n by

$$\log \varphi_n(u) = \frac{1}{n} \log \varphi(u).$$

Since dividing a Lévy–Khintchine exponent by n only rescales its parameters, $\log \varphi_n(u)$ is again of Lévy–Khintchine form (15). Hence by assumption $\varphi_n(u)$ is a characteristic function. Moreover,

$$\varphi(u) = (\varphi_n(u))^n.$$

Therefore the distribution corresponding to φ can be written, for every $n \in \mathbb{N}$, as the distribution of a sum of n independent identically distributed random variables. Hence it is infinitely divisible by Definition 5.1.

We now prove the converse implication. Assume that X is infinitely divisible and let $\varphi(u)$ be the characteristic function of X . By infinite divisibility, for every n , there exist i.i.d. random variables $X_1^{(n)}, \dots, X_n^{(n)}$ such that $X \stackrel{d}{=} X_1^{(n)} + \dots + X_n^{(n)}$. If φ is the characteristic function of X , and φ_n is the characteristic function of $X_1^{(n)}$, then

$$\varphi(u) = \varphi_n(u)^n.$$

By Proposition 5.6, $\varphi_n(u) \rightarrow 1$ for every u . We recall from Remark 3.2 that this convergence is uniform on bounded intervals of u , and denote this by $\varphi_n(u) \xrightarrow{uc} 1$.

Thus, for every bounded interval $K \subset \mathbb{R}$,

$$\sup_{u \in K} |\varphi_n(u) - 1| \rightarrow 0.$$

Hence, for all sufficiently large n , $\varphi_n(u)$ lies in a fixed neighbourhood of 1, uniformly for $u \in K$. On this neighbourhood we may choose one branch of the complex logarithm. Therefore $\log \varphi_n(u)$ is well-defined on K for large n , and Taylor expansion of $\log z$ around $z = 1$ gives

$$\log \varphi_n(u) = (\varphi_n(u) - 1)(1 + \varepsilon_n(u)),$$

where $\varepsilon_n(u) \xrightarrow{uc} 0$. Consequently, $\log \varphi(u) = n(\varphi_n(u) - 1)(1 + \varepsilon_n(u))$.

Moreover, $\varphi(u) \neq 0$ for all u . Indeed, if $\varphi(u) = 0$, then $\varphi_n(u)^n = 0$, so $\varphi_n(u) = 0$ for every n , contradicting $\varphi_n(u) \rightarrow 1$. Hence a continuous logarithm of φ can be chosen.

Let F_n denote the law of $X_1^{(n)}$ on $\mathbb{R}$. From the previous expansion we have

$$\log \varphi(u) = (1 + \varepsilon_n(u)) \int (e^{iux} - 1) n F_n(dx).$$

It is therefore natural to introduce the measure μ_n by $\mu_n(B) = n F_n(B)$, so that $\mu_n(dx) = n F_n(dx)$. With this notation, $\log \varphi(u) = (1 + \varepsilon_n(u)) \int (e^{iux} - 1) \mu_n(dx)$.

The measure μ_n has a useful interpretation. It records the expected number of summands in the n -th row whose values fall in a given set. Indeed, if

$$H_n(B) = \sum_{k=1}^n \mathbf{1}_{\{X_k^{(n)} \in B\}},$$

then $H_n(B)$ counts how many of the summands $X_1^{(n)}, \dots, X_n^{(n)}$ take values in B . Hence

$$\mathbb{E}[H_n(B)] = \sum_{k=1}^n \mathbb{P}(X_k^{(n)} \in B) = n F_n(B) = \mu_n(B). \quad (16)$$

Thus, $\mu_n(B)$ can be interpreted as the expected number of summands in the n -th row whose values fall in B . This also explains why the behaviour of μ_n near the origin has to be treated separately from its behaviour away from the origin. Since $\mu_n(\mathbb{R}) = n$, the total mass of μ_n tends to infinity as $n \rightarrow \infty$, so the measures μ_n cannot converge as finite measures on all of $\mathbb{R}$. However, by Proposition 5.6, the individual summands $X_k^{(n)}$ converge in distribution to zero. Hence the diverging mass is expected to accumulate near the origin.

Away from zero, the summands are not negligible. They correspond to jumps of a size that may influence the limit. Hence the relevant question is whether the expected number of such jumps remains bounded. In terms of μ_n , this amounts to controlling $\mu_n(N^c)$ for neighbourhoods N of zero. The next step is therefore to show that, although $\mu_n(\mathbb{R}) \rightarrow \infty$, the measures μ_n remain controlled away from zero.

We shall use the following auxiliary estimate, whose proof is given in Appendix A.5. For every $a > 0$,

$$\limsup_{n \rightarrow \infty} \mu_n([-a, a]^c) \leq C a \int_0^{1/a} |\Re \log \varphi(v)| dv, \quad (17)$$

where $C > 0$ is a constant and φ is the characteristic function of X . In particular, the right-hand side is finite for each fixed $a > 0$. Hence the measures μ_n have uniformly bounded mass outside every neighbourhood of the origin.

We now prove the corresponding estimate near the origin. Namely, we show that

$$\limsup_{n \rightarrow \infty} \int_{[-1,1]} x^2 \mu_n(dx) < \infty. \quad (18)$$

The idea is to compare x^2 with $1 - \cos x$ near the origin, since the latter quantity is directly related to the real part of the characteristic function.

For $|x| \leq 1$, Taylor's formula with Lagrange remainder gives $\cos x = 1 - \frac{x^2}{2} \cos(\theta_x)$, where θ_x lies between 0 and x . Hence $|\theta_x| \leq 1$, so $\cos(\theta_x) \geq \cos(1) > 0$. Thus, with $\xi = \cos(1)/2$,

$$1 - \cos x \geq \xi x^2, \quad |x| \leq 1.$$

It follows that

$$\xi \int_{[-1,1]} x^2 \mu_n(dx) \leq \int_{[-1,1]} (1 - \cos x) \mu_n(dx) \leq \int_{\mathbb{R}} (1 - \cos x) \mu_n(dx).$$

Using $\mu_n(dx) = n F_n(dx)$ and

$$\Re \varphi_n(1) = \Re \int_{\mathbb{R}} e^{ix} F_n(dx) = \int_{\mathbb{R}} \cos x F_n(dx),$$

we obtain

$$\int_{\mathbb{R}} (1 - \cos x) \mu_n(dx) = n \int_{\mathbb{R}} (1 - \cos x) F_n(dx) = n(1 - \Re \varphi_n(1)).$$

Therefore

$$\int_{[-1,1]} x^2 \mu_n(dx) \leq \frac{1}{\xi} n(1 - \Re \varphi_n(1)).$$

It remains only to show that the right-hand side is bounded as $n \rightarrow \infty$.

From

$$\log \varphi(u) = n(\varphi_n(u) - 1)(1 + \varepsilon_n(u)), \quad \varepsilon_n(u) \rightarrow 0,$$

we obtain, by setting $u = 1$,

$$n(\varphi_n(1) - 1) = \frac{\log \varphi(1)}{1 + \varepsilon_n(1)} \rightarrow \log \varphi(1).$$

Taking real parts gives $n(1 - \Re \varphi_n(1)) \rightarrow -\Re \log \varphi(1)$. This limit is finite, since $\varphi(1) \neq 0$. Hence

$$\limsup_{n \rightarrow \infty} \int_{[-1,1]} x^2 \mu_n(dx) \leq \frac{1}{\xi} \lim_{n \rightarrow \infty} n(1 - \Re \varphi_n(1)) < \infty,$$

which proves the claim. We have now obtained the two estimates needed to control μ_n . Away from the origin, the mass $\mu_n(N^c)$ remains bounded, while near the origin the weighted mass $\int_{[-1,1]} x^2 \mu_n(dx)$ remains bounded. Thus the only possible accumulation of mass occurs at zero, and it is controlled after multiplication by x^2 .

These two estimates suggest introducing a measure which weights μ_n by a factor that behaves like x^2 near zero and like 1 away from zero. Consider $h(x) = \frac{x^2}{1+x^2}$. This function is bounded, behaves like x^2 near the origin, and is therefore compatible with both (18) and (17). Define

$$\alpha_n = \int_{\mathbb{R}} \frac{y^2}{1+y^2} \mu_n(dy).$$

Splitting the integral at $[-1, 1]$, we get

$$\alpha_n = \int_{[-1,1]} \frac{y^2}{1+y^2} \mu_n(dy) + \int_{[-1,1]^c} \frac{y^2}{1+y^2} \mu_n(dy).$$

Since $\frac{y^2}{1+y^2} \leq y^2$ on $[-1, 1]$ and $\frac{y^2}{1+y^2} \leq 1$ on $[-1, 1]^c$, it follows that

$$\alpha_n \leq \int_{[-1,1]} y^2 \mu_n(dy) + \mu_n([-1, 1]^c).$$

By (18) and (17), both terms on the right-hand side are bounded in $\limsup$. Hence $\limsup_{n \rightarrow \infty} \alpha_n < \infty$.

Thus the weighted measures have bounded total mass. Whenever $\alpha_n > 0$, we normalize this mass and define a probability measure Q_n by

$$Q_n(B) = \frac{1}{\alpha_n} \int_B \frac{y^2}{1+y^2} \mu_n(dy), \quad B \in \mathcal{B}(\mathbb{R}).$$

Let G_n denote the corresponding distribution function, so that $G_n(x) = Q_n((-\infty, x])$. Equivalently, $Q_n(dx) = \frac{1}{\alpha_n} \frac{x^2}{1+x^2} \mu_n(dx)$. Using this change of measure, we can write

$$\log \varphi(u) = (1 + \varepsilon_n(u)) \alpha_n \int_{\mathbb{R}} (e^{iux} - 1) \frac{1+x^2}{x^2} Q_n(dx).$$

The problem is that this integrand is not bounded near the origin. Even if G_n converged in distribution, this would not by itself guarantee convergence of the integral, since convergence in distribution only guarantees convergence of integrals when the integrand is bounded and continuous; see Le Gall (2022, Def. 10.9). To remove this local singularity, we subtract and add the same linear term:

$$e^{iux} - 1 = \left(e^{iux} - 1 - \frac{iux}{1+x^2} \right) + \frac{iux}{1+x^2}.$$

Writing $\beta_n = \int \frac{x}{1+x^2} \mu_n(dx)$, this becomes

$$\log \varphi(u) = (1 + \varepsilon_n(u)) \alpha_n \int_{\mathbb{R}} \left(e^{iux} - 1 - \frac{iux}{1+x^2} \right) \frac{1+x^2}{x^2} Q_n(dx) + i(1 + \varepsilon_n(u)) \beta_n u. \quad (19)$$

Because (α_n) is a bounded sequence, the Bolzano–Weierstrass theorem states that there exists a subsequence $(\alpha_{n'})$ such that $\alpha_{n'} \rightarrow \alpha \geq 0$.

If $\alpha = 0$, the integral term disappears in the limit, and the limiting characteristic function is of the form $\varphi(u) = e^{i\beta u}$. Since X is non-degenerate, $\alpha > 0$. For $\alpha_{n'} > 0$, consider the probability measures $Q_{n'}$ defined above, and let $G_{n'}$ be their distribution functions. By the bounds (18) and (17), $(G_{n'})$ is mass-preserving. Hence, by Proposition 5.3, we may pass to a further subsequence $(G_{\bar{n}})$, where $\bar{n}$ is a subsequence of n' , such that $G_{\bar{n}} \xrightarrow{D} G$ for some distribution function G . Let Q denote the probability measure corresponding to G .

For $x \neq 0$, define

$$\eta(x, u) = \left(e^{iux} - 1 - \frac{iux}{1+x^2} \right) \frac{1+x^2}{x^2}.$$

This is the integrand with respect to $Q_{\bar{n}}$ in (19). Although its definition contains the factor x^{-2} , the singularity at $x = 0$ is removable. Defining

$$\eta(0, u) = \lim_{x \rightarrow 0} \eta(x, u) = -\frac{u^2}{2},$$

the function $\eta(\cdot, u)$ becomes bounded and continuous. Therefore $G_{\bar{n}} \xrightarrow{D} G$, together with the defining property of convergence in distribution, gives $\int \eta(x, u) Q_{\bar{n}}(dx) \rightarrow \int \eta(x, u) Q(dx)$.

Thus the integral term converges. We now argue that the individual sequences appearing in (19) converge. Since $\bar{n}$ is a subsequence of n' , we still have $\alpha_{\bar{n}} \rightarrow \alpha$. Since also $\log \varphi(u)$ is fixed and $\varepsilon_{\bar{n}}(u) \rightarrow 0$, the remaining term $i(1 + \varepsilon_{\bar{n}}(u)) \beta_{\bar{n}} u$ is forced to converge as well. Hence, along this subsequence, $\beta_{\bar{n}} \rightarrow \beta$, and we obtain

$$\log \varphi(u) = \alpha \int \eta(x, u) Q(dx) + i\beta u.$$

Suppose $Q(\{0\}) > 0$ and let $\sigma^2 = \alpha Q(\{0\})$. Then $\log \varphi(u) = i\beta u - \frac{\sigma^2 u^2}{2} + \alpha \int_{\{0\}^c} \eta(x, u) Q(dx)$. $\square$

A natural extension of Theorem 5.5 is to ask whether (15) always defines a characteristic function, and whether the representation is unique. Both statements are true, see Breiman (1992, Prop 9.21 and the surrounding discussion).

In particular, if X is infinitely divisible and its characteristic function is written in the form (15), then the constant β , the coefficient σ^2 , and the finite measure ν are uniquely determined.

A Appendix

A.1 Proof that (1.2) implies (1.1)

We show that Definition 1.2 implies Definition 1.1. Assume that X is non-degenerate and stable in the sense of Definition 1.2. Thus, for every $n > 1$, there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$X_1 + \cdots + X_n \stackrel{d}{=} c_n X + d_n,$$

where $X_1, \dots, X_n$ are independent copies of X . We also use the standard fact that the norming constants are of the form

$$c_n = n^{1/\alpha}$$

for some $\alpha > 0$.

Let $m, k \in \mathbb{N}$, and set $N = m + k$. We split the sum with N terms into two independent blocks:

$$X_1 + \cdots + X_N = (X_1 + \cdots + X_m) + (X_{m+1} + \cdots + X_{m+k}).$$

By Definition 1.2, the first block has the same distribution as $m^{1/\alpha}Y + d_m$, where Y is a copy of X . Similarly, the second block has the same distribution as $k^{1/\alpha}Z + d_k$, where Z is an independent copy of X . Since the two blocks are independent, we obtain

$$X_1 + \cdots + X_N \stackrel{d}{=} m^{1/\alpha}Y + k^{1/\alpha}Z + d_m + d_k.$$

On the other hand, applying Definition 1.2 directly to the whole sum gives

$$X_1 + \cdots + X_N \stackrel{d}{=} N^{1/\alpha}X + d_N.$$

Hence

$$m^{1/\alpha}Y + k^{1/\alpha}Z + d_m + d_k \stackrel{d}{=} N^{1/\alpha}X + d_N.$$

Since $N = m + k$, division by $N^{1/\alpha} = (m + k)^{1/\alpha}$ gives

$$\left(\frac{m}{m+k}\right)^{1/\alpha} Y + \left(\frac{k}{m+k}\right)^{1/\alpha} Z \stackrel{d}{=} X + r_{m,k},$$

where

$$r_{m,k} := \frac{d_{m+k} - d_m - d_k}{(m+k)^{1/\alpha}}.$$

This identity holds for every $m, k \in \mathbb{N}$. This is the point at which Definition 1.2 has been used: it gives a two-block representation for every choice of block sizes m and k .

Now fix arbitrary $a, b > 0$. We want to prove that there exist constants $C > 0$ and $D \in \mathbb{R}$ such that

$$aY + bZ \stackrel{d}{=} CX + D.$$

Define

$$C = (a^\alpha + b^\alpha)^{1/\alpha}$$

and

$$p = \left(\frac{a}{C}\right)^\alpha = \frac{a^\alpha}{a^\alpha + b^\alpha}.$$

Then $p \in (0, 1)$, and

$$1 - p = \left(\frac{b}{C}\right)^\alpha.$$

Equivalently,

$$p^{1/\alpha} = \frac{a}{C} \quad \text{and} \quad (1-p)^{1/\alpha} = \frac{b}{C}.$$

We now choose a sequence of integer pairs (m_j, k_j) such that

$$\frac{m_j}{m_j + k_j} \rightarrow p \quad \text{as } j \rightarrow \infty.$$

This is possible because the identity above holds for every pair $m, k \in \mathbb{N}$. For example, set $N_j = j$, $m_j = \lfloor pN_j \rfloor$, and $k_j = N_j - m_j$. Then $m_j + k_j = N_j$, and

$$0 \leq p - \frac{m_j}{N_j} < \frac{1}{N_j}.$$

Since $N_j = j \rightarrow \infty$, it follows that

$$\frac{m_j}{m_j + k_j} = \frac{m_j}{N_j} \rightarrow p.$$

Consequently,

$$\frac{k_j}{m_j + k_j} \rightarrow 1 - p.$$

For each j , we may insert $m = m_j$ and $k = k_j$ into the identity obtained above. Thus

$$\left(\frac{m_j}{m_j + k_j} \right)^{1/\alpha} Y + \left(\frac{k_j}{m_j + k_j} \right)^{1/\alpha} Z \stackrel{d}{=} X + r_j,$$

where $r_j := r_{m_j, k_j}$. Since

$$\left(\frac{m_j}{m_j + k_j} \right)^{1/\alpha} \rightarrow p^{1/\alpha} = \frac{a}{C}$$

and

$$\left(\frac{k_j}{m_j + k_j} \right)^{1/\alpha} \rightarrow (1 - p)^{1/\alpha} = \frac{b}{C},$$

the left-hand side converges in distribution to

$$\frac{a}{C}Y + \frac{b}{C}Z.$$

Therefore

$$X + r_j \xrightarrow{d} \frac{a}{C}Y + \frac{b}{C}Z \quad \text{as } j \rightarrow \infty.$$

By the convergence of types theorem, since X is non-degenerate and $X + r_j$ converges in distribution, there exists $r \in \mathbb{R}$ such that $r_j \rightarrow r$. Hence

$$\frac{a}{C}Y + \frac{b}{C}Z \stackrel{d}{=} X + r.$$

Multiplying by C , we obtain

$$aY + bZ \stackrel{d}{=} CX + Cr.$$

Setting $D = Cr$, this becomes

$$aY + bZ \stackrel{d}{=} CX + D.$$

Since $a, b > 0$ were arbitrary, this proves Definition 1.1.

A.2 Proof of Equation 3

We now show that the norming constants of a stable distribution must be of the form $c_n = n^{1/\alpha}$. The argument follows the standard idea of writing sums of independent copies in different ways and comparing the corresponding normalising constants.

Let X be a non-degenerate stable random variable, and let $X_1, X_2, \dots$ be independent copies of X . By Definition 1.2, for each $n \in \mathbb{N}$, there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$X_1 + \dots + X_n \stackrel{d}{=} c_n X + d_n.$$

Since translations do not affect the norming constants, we may first symmetrise. Let X' be an independent copy of X , and put

$$\tilde{X} = X - X'.$$

Then $\tilde{X}$ is symmetric and non-degenerate. Moreover, if $\tilde{X}_1, \dots, \tilde{X}_n$ are independent copies of $\tilde{X}$, then

$$\tilde{X}_1 + \dots + \tilde{X}_n \stackrel{d}{=} c_n \tilde{X}.$$

Indeed, the translation terms cancel:

$$(S_n - S'_n) \stackrel{d}{=} (c_n X + d_n) - (c_n X' + d_n) = c_n (X - X').$$

It is therefore enough to prove the result in the symmetric strictly stable case. Thus, from now on, we assume that X is symmetric, non-degenerate, and that

$$X_1 + \dots + X_n \stackrel{d}{=} c_n X.$$

We first derive a multiplicative relation for the constants. Consider the sum of mn independent copies of X . Directly from stability,

$$X_1 + \dots + X_{mn} \stackrel{d}{=} c_{mn} X.$$

On the other hand, we can split the same sum into m blocks of length n :

$$X_1 + \dots + X_{mn} = \sum_{j=1}^m (X_{(j-1)n+1} + \dots + X_{jn}).$$

Each block has distribution $c_n X$, and the blocks are independent. Hence

$$X_1 + \dots + X_{mn} \stackrel{d}{=} c_n (X_1 + \dots + X_m).$$

Applying stability once more gives

$$c_n (X_1 + \dots + X_m) \stackrel{d}{=} c_n c_m X.$$

Therefore

$$c_{mn} X \stackrel{d}{=} c_m c_n X.$$

Since X is non-degenerate, this implies

$$c_{mn} = c_m c_n.$$

In particular, for every fixed integer $r \geq 2$,

$$c_{r^j} = c_r^j, \quad j \in \mathbb{N}.$$

Since X is non-degenerate, $c_r > 1$ for $r \geq 2$. Hence there is a unique number $\alpha_r > 0$ such that

$$c_r = r^{1/\alpha_r}.$$

It follows that

$$c_{r^j} = c_r^j = \left(r^{1/\alpha_r}\right)^j = (r^j)^{1/\alpha_r}.$$

Thus, along the subsequence $n = r^j$, we have

$$c_n = n^{1/\alpha_r}.$$

It remains to show that the exponent α_r does not depend on the chosen base r . For this we need one further estimate. From stability,

$$X_1 + \dots + X_{m+n} \stackrel{d}{=} c_{m+n} X.$$

Splitting the sum into two independent blocks of lengths m and n , we get

$$c_{m+n} X \stackrel{d}{=} c_m X_1 + c_n X_2.$$

Equivalently,

$$X \stackrel{d}{=} \frac{c_m}{c_{m+n}} X_1 + \frac{c_n}{c_{m+n}} X_2.$$

Let $x > 0$. If

$$X_1 \geq 0 \quad \text{and} \quad X_2 > \frac{c_{m+n}}{c_n} x,$$

then

$$\frac{c_m}{c_{m+n}} X_1 + \frac{c_n}{c_{m+n}} X_2 > x.$$

Hence, by independence and symmetry,

$$\mathbb{P}(X > x) \geq \mathbb{P}(X_1 \geq 0) \mathbb{P}\left(X_2 > \frac{c_{m+n}}{c_n} x\right) \geq \frac{1}{2} \mathbb{P}\left(X > \frac{c_{m+n}}{c_n} x\right).$$

This implies that the ratios c_n/c_{m+n} are bounded. Indeed, if this were not true, we could find m_k, n_k such that

$$\frac{c_{n_k}}{c_{m_k+n_k}} \rightarrow \infty, \quad \text{or equivalently} \quad \frac{c_{m_k+n_k}}{c_{n_k}} \rightarrow 0.$$

Then, for fixed $x > 0$,

$$\mathbb{P}(X > x) \geq \frac{1}{2} \mathbb{P}\left(X > \frac{c_{m_k+n_k}}{c_{n_k}} x\right) \rightarrow \frac{1}{2} \mathbb{P}(X > 0).$$

Since X is symmetric and non-degenerate, $\mathbb{P}(X > 0) > 0$. This would give a positive lower bound for $\mathbb{P}(X > x)$ for arbitrarily large x , which contradicts $\mathbb{P}(X > x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore there exists a constant $K < \infty$ such that

$$\frac{c_n}{c_k} \leq K \quad \text{whenever } k > n.$$

We now compare two arbitrary bases. Let $r, \rho \geq 2$ be integers, and write

$$c_r = r^{1/\alpha}, \quad c_\rho = \rho^{1/\beta}.$$

Then

$$c_{r^j} = (r^j)^{1/\alpha}, \quad c_{\rho^k} = (\rho^k)^{1/\beta}.$$

We show that $\alpha = \beta$.

Let

$$m = \rho^k.$$

Choose j such that

$$r^j \leq m < r^{j+1},$$

and put

$$n = r^j.$$

Then

$$n \leq m < rn.$$

Thus

$$c_m = m^{1/\beta} \leq (rn)^{1/\beta} = r^{1/\beta} n^{1/\beta}.$$

Since $c_n = n^{1/\alpha}$, we have $n = c_n^\alpha$, and therefore

$$n^{1/\beta} = c_n^{\alpha/\beta}.$$

Hence

$$c_m \leq r^{1/\beta} c_n^{\alpha/\beta}.$$

Dividing by c_n , we obtain

$$\frac{c_m}{c_n} \leq r^{1/\beta} c_n^{\alpha/\beta-1}.$$

On the other hand, since $m \geq n$, the bounded-ratio estimate gives

$$\frac{c_n}{c_m} \leq K,$$

and therefore

$$\frac{c_m}{c_n} \geq \frac{1}{K}.$$

Combining the two inequalities gives

$$\frac{1}{K} \leq r^{1/\beta} c_n^{\alpha/\beta-1}.$$

Now let $k \rightarrow \infty$. Then $m = \rho^k \rightarrow \infty$, and since $n \leq m < rn$, also $n \rightarrow \infty$. Hence

$$c_n = n^{1/\alpha} \rightarrow \infty.$$

If $\beta > \alpha$, then

$$\frac{\alpha}{\beta} - 1 < 0,$$

so

$$c_n^{\alpha/\beta-1} \rightarrow 0.$$

The right-hand side would therefore tend to 0, while the left-hand side is the fixed positive number $1/K$. This is impossible. Hence

$$\beta \leq \alpha.$$

By reversing the roles of r and ρ , the same argument gives

$$\alpha \leq \beta.$$

Consequently,

$$\alpha = \beta.$$

Thus the exponent obtained from the subsequence $r, r^2, r^3, \dots$ is the same for every base $r \geq 2$. Denote this common exponent by α . Taking $r = n$ gives, for every $n \geq 2$,

$$c_n = n^{1/\alpha}.$$

Since $c_1 = 1$, the formula holds for all $n \in \mathbb{N}$:

$$c_n = n^{1/\alpha}.$$

Finally, one has $0 < \alpha \leq 2$. The positivity follows from $c_r > 1$. The upper bound is obtained from a tail argument: stability implies that the tails of X cannot be thinner than those corresponding to the normalisation $n^{1/\alpha}$. If $\alpha > 2$, this tail estimate would imply $\mathbb{E}[X^2] < \infty$. But then taking variances in

$$X_1 + \dots + X_n \stackrel{d}{=} c_n X$$

would give

$$n \operatorname{Var}(X) = c_n^2 \operatorname{Var}(X),$$

and hence $c_n = \sqrt{n}$, corresponding to $\alpha = 2$. Therefore $\alpha > 2$ is impossible. The Gaussian case corresponds to $\alpha = 2$; excluding it gives $0 < \alpha < 2$.

A.3 Proof of equivalence between Definition 1.3 and Definition 1.4

We first show that the representation in Definition 1.3 can be written in the form of the 1-parametrization. Let $Z_{\alpha,\beta}$ denote the standard stable random variable in Definition 1.3, and assume that

$$X \stackrel{d}{=} aZ_{\alpha,\beta} + b, \quad a \neq 0.$$

Set $\gamma = |a|$. If $a > 0$, define $\tilde{\beta} = \beta$, while if $a < 0$, define $\tilde{\beta} = -\beta$. In the case $a < 0$, we use that $-Z_{\alpha,\beta} \stackrel{d}{=} Z_{\alpha,-\beta}$, which follows directly from the characteristic function by replacing u by $-u$. Hence, in both cases,

$$aZ_{\alpha,\beta} \stackrel{d}{=} \gamma Z_{\alpha,\tilde{\beta}}.$$

For $\alpha \neq 1$, choose $\delta = b$. Then

$$X \stackrel{d}{=} \gamma Z_{\alpha,\tilde{\beta}} + \delta,$$

which is exactly the form in Definition 1.4. For $\alpha = 1$, choose

$$\delta = b - \tilde{\beta} \frac{2}{\pi} \gamma \log \gamma.$$

Then

$$X \stackrel{d}{=} \gamma Z_{1,\tilde{\beta}} + \left(\delta + \tilde{\beta} \frac{2}{\pi} \gamma \log \gamma \right),$$

which is the $\alpha = 1$ case of Definition 1.4.

Conversely, assume that $X \sim \mathbf{S}(\alpha, \beta, \gamma, \delta; 1)$. We show that X can be written in the form $aZ_{\alpha,\beta} + b$. If $\alpha \neq 1$, then by definition

$$X \stackrel{d}{=} \gamma Z_{\alpha,\beta} + \delta,$$

so we may take $a = \gamma$ and $b = \delta$. If $\alpha = 1$, then

$$X \stackrel{d}{=} \gamma Z_{1,\beta} + \left(\delta + \beta \frac{2}{\pi} \gamma \log \gamma \right),$$

so we may take

$$a = \gamma, \quad b = \delta + \beta \frac{2}{\pi} \gamma \log \gamma.$$

Thus every random variable in the 1-parametrization is also of the form given in Definition 1.3. Hence the two definitions describe the same class of distributions.

The extra term in the case $\alpha = 1$ appears because scaling gives $\log |\gamma u| = \log \gamma + \log |u|$ in the characteristic function.

A.4 Proof of Equation 12

Set

$$\theta := \phi + \frac{\pi}{2}, \quad 0 < \theta < \pi.$$

Then $\phi = \theta - \frac{\pi}{2}$, and we define the reparametrization of Γ_2 by

$$z(\theta) := \rho \left(\theta - \frac{\pi}{2} \right) e^{i(\theta - \pi/2)} = -i \rho \left(\theta - \frac{\pi}{2} \right) e^{i\theta}, \quad 0 < \theta < \pi.$$

Hence we get the integral

$$\Re \int_{\Gamma_2} v(z) dz = \Re \int_0^\pi v(z(\theta)) z'(\theta) d\theta = \int_0^\pi \Re(v(z(\theta)) z'(\theta)) d\theta.$$

We want to separate the real and imaginary parts of $v(z(\theta))z'(\theta)$. Recall that

$$v(z) = \frac{1}{\pi} x^{-2} i z \exp \left(i z - z^\alpha x^{-\alpha} e^{i\pi\alpha/2} \right).$$

Let

$$E(z) := i z - z^\alpha x^{-\alpha} e^{i\pi\alpha/2}$$

denote the exponent. For notational convenience, write

$$r(\theta) := \rho\left(\theta - \frac{\pi}{2}\right).$$

Then the parametrization of Γ_2 is

$$z(\theta) = -ir(\theta)e^{i\theta}.$$

We first show that along Γ_2 ,

$$\Im(E(z(\theta))) = 0.$$

Since

$$z(\theta) = -ir(\theta)e^{i\theta},$$

we have

$$iz(\theta) = r(\theta)e^{i\theta},$$

and

$$z(\theta)^\alpha = (-ir(\theta)e^{i\theta})^\alpha = r(\theta)^\alpha e^{-i\pi\alpha/2} e^{i\alpha\theta}.$$

Therefore

$$x^{-\alpha} e^{i\pi\alpha/2} z(\theta)^\alpha = x^{-\alpha} r(\theta)^\alpha e^{i\alpha\theta}.$$

Substituting these expressions into E , we obtain

$$E(\theta) := E(z(\theta)) = r(\theta)e^{i\theta} - x^{-\alpha} r(\theta)^\alpha e^{i\alpha\theta}.$$

Hence

$$E(\theta) = (r(\theta) \cos \theta - x^{-\alpha} r(\theta)^\alpha \cos(\alpha\theta)) + i(r(\theta) \sin \theta - x^{-\alpha} r(\theta)^\alpha \sin(\alpha\theta)).$$

Thus

$$\Re E(\theta) = r(\theta) \cos \theta - x^{-\alpha} r(\theta)^\alpha \cos(\alpha\theta),$$

and

$$\Im E(\theta) = r(\theta) \sin \theta - x^{-\alpha} r(\theta)^\alpha \sin(\alpha\theta). \quad (20)$$

From the parametrization of Γ_2 , we have

$$r(\theta)^{1-\alpha} = x^{-\alpha} \frac{\sin(\alpha\theta)}{\sin \theta}.$$

Multiplying both sides by $r(\theta)^\alpha \sin \theta$, this gives

$$r(\theta) \sin \theta = x^{-\alpha} r(\theta)^\alpha \sin(\alpha\theta).$$

Substituting this identity into (20), we obtain

$$\Im E(\theta) = 0.$$

Therefore $E(\theta)$ is real-valued along Γ_2 , and hence

$$E(\theta) = \Re E(\theta).$$

Next, we rewrite the real part of $E(\theta)$ in a more convenient form. From

$$r(\theta)^{1-\alpha} = x^{-\alpha} \frac{\sin(\alpha\theta)}{\sin \theta},$$

we obtain

$$x^{-\alpha} r(\theta)^\alpha = r(\theta) \frac{\sin \theta}{\sin(\alpha\theta)}.$$

Substituting this into $\Re E(\theta)$ yields

$$\Re E(\theta) = r(\theta) \left(\cos \theta - \frac{\sin \theta \cos(\alpha\theta)}{\sin(\alpha\theta)} \right).$$

Combining the terms gives

$$\Re E(\theta) = r(\theta) \frac{\cos \theta \sin(\alpha\theta) - \sin \theta \cos(\alpha\theta)}{\sin(\alpha\theta)}.$$

Using the identity

$$\sin A \cos B - \cos A \sin B = \sin(A - B),$$

we get

$$\cos \theta \sin(\alpha\theta) - \sin \theta \cos(\alpha\theta) = \sin((\alpha - 1)\theta) = -\sin((1 - \alpha)\theta).$$

Therefore

$$\Re E(\theta) = -r(\theta) \frac{\sin((1 - \alpha)\theta)}{\sin(\alpha\theta)}.$$

Now, from

$$r(\theta)^{1-\alpha} = x^{-\alpha} \frac{\sin(\alpha\theta)}{\sin \theta},$$

we have

$$r(\theta) = x^{\alpha/(\alpha-1)} \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{1/(1-\alpha)}.$$

Substituting this into the expression for $\Re E(\theta)$, we obtain

$$\Re E(\theta) = -x^{\alpha/(\alpha-1)} \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{1/(1-\alpha)} \frac{\sin((1 - \alpha)\theta)}{\sin(\alpha\theta)}.$$

Hence

$$\Re E(\theta) = -x^{\alpha/(\alpha-1)} a(\theta),$$

where

$$a(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{1/(1-\alpha)} \frac{\sin((1 - \alpha)\theta)}{\sin(\alpha\theta)}.$$

Since $E(\theta)$ is real-valued along Γ_2 , we have

$$\exp(E(\theta)) = \exp(\Re E(\theta)) = \exp\left(-x^{\alpha/(\alpha-1)} a(\theta)\right).$$

Therefore

$$\Re(v(z(\theta))z'(\theta)) = \pi^{-1} x^{-2} \exp\left(-x^{\alpha/(\alpha-1)} a(\theta)\right) \Re(iz(\theta)z'(\theta)).$$

It remains to compute $\Re(iz(\theta)z'(\theta))$. Since

$$z(\theta) = -ir(\theta)e^{i\theta},$$

we have

$$z'(\theta) = -ie^{i\theta} (r'(\theta) + ir(\theta)).$$

Moreover,

$$iz(\theta) = r(\theta)e^{i\theta}.$$

Therefore

$$\begin{aligned} iz(\theta)z'(\theta) &= r(\theta)e^{i\theta} [-ie^{i\theta} (r'(\theta) + ir(\theta))] \\ &= (r(\theta)^2 - ir(\theta)r'(\theta)) e^{2i\theta}. \end{aligned}$$

Taking real parts gives

$$\Re(iz(\theta)z'(\theta)) = r(\theta)^2 \cos(2\theta) + r(\theta)r'(\theta) \sin(2\theta).$$

Equivalently,

$$\Re(iz(\theta)z'(\theta)) = r(\theta)^2 \left(\cos(2\theta) + \frac{r'(\theta)}{r(\theta)} \sin(2\theta) \right).$$

We now compute $r'(\theta)/r(\theta)$. Since

$$r(\theta) = x^{\alpha/(\alpha-1)} \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{1/(1-\alpha)},$$

logarithmic differentiation gives

$$\frac{r'(\theta)}{r(\theta)} = \frac{1}{1-\alpha} (\alpha \cot(\alpha\theta) - \cot \theta).$$

Thus

$$\Re(iz(\theta)z'(\theta)) = r(\theta)^2 \left[\cos(2\theta) + \frac{\sin(2\theta)}{1-\alpha} (\alpha \cot(\alpha\theta) - \cot \theta) \right].$$

We simplify the expression in brackets. Using

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

and the definitions of cotangent, we get

$$\begin{aligned} & \cos(2\theta) + \frac{\sin(2\theta)}{1-\alpha} (\alpha \cot(\alpha\theta) - \cot \theta) \\ &= \cos(2\theta) + \frac{2 \sin \theta \cos \theta}{1-\alpha} \left(\alpha \frac{\cos(\alpha\theta)}{\sin(\alpha\theta)} - \frac{\cos \theta}{\sin \theta} \right) \\ &= \cos(2\theta) + \frac{2\alpha \sin \theta \cos \theta \cos(\alpha\theta)}{(1-\alpha) \sin(\alpha\theta)} - \frac{2 \cos^2 \theta}{1-\alpha}. \end{aligned}$$

This can be rewritten as

$$\frac{2\alpha \cos \theta \sin((1-\alpha)\theta)}{(1-\alpha) \sin(\alpha\theta)} - 1.$$

Indeed, this follows from

$$\sin((1-\alpha)\theta) = \sin \theta \cos(\alpha\theta) - \cos \theta \sin(\alpha\theta).$$

Since

$$r(\theta)^2 = x^{2\alpha/(\alpha-1)} \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{2/(1-\alpha)},$$

we conclude that

$$\Re(iz(\theta)z'(\theta)) = x^{2\alpha/(\alpha-1)} b(\theta),$$

where

$$b(\theta) = \left(\frac{\sin(\alpha\theta)}{\sin \theta} \right)^{2/(1-\alpha)} \left(\frac{2\alpha \cos \theta \sin((1-\alpha)\theta)}{(1-\alpha) \sin(\alpha\theta)} - 1 \right).$$

Combining the above identities, we get

$$\begin{aligned} \Re \int_{\Gamma_2} v(z) dz &= \pi^{-1} x^{-2} x^{2\alpha/(\alpha-1)} \int_0^\pi \exp\{-x^{\alpha/(\alpha-1)} a(\theta)\} b(\theta) d\theta \\ &= \pi^{-1} x^{2/(\alpha-1)} \int_0^\pi \exp\{-x^{\alpha/(\alpha-1)} a(\theta)\} b(\theta) d\theta. \end{aligned}$$

A.5 Proof of Equation 17

Proof. We start by using the inequality given in (Prop. 8.29, Breiman, 1992). Since $\mu_n = nF_n$, we obtain

$$\mu_n([-a, a]^c) \leq \alpha a \int_0^{1/a} n(1 - \Re f_n(v)) dv.$$

We want to understand the limiting behaviour of the integrand

$$n(1 - \Re f_n(v)).$$

We have

$$\log f(v) = n(f_n(v) - 1)(1 + \varepsilon_n(v)),$$

and hence

$$n(f_n(v) - 1) = \frac{\log f(v)}{1 + \varepsilon_n(v)}.$$

Since $\varepsilon_n(v) \rightarrow 0$, it follows pointwise that

$$n(f_n(v) - 1) \rightarrow \log f(v).$$

Taking real parts gives

$$n(\Re f_n(v) - 1) \rightarrow \Re \log f(v),$$

or equivalently

$$n(1 - \Re f_n(v)) \rightarrow -\Re \log f(v).$$

We now use the bounded convergence theorem from LIM Schilling (2017, Exc. 22.7). The pointwise convergence has just been shown for every $v \in [0, 1/a]$. It remains to note that the integrand is uniformly bounded on this interval. Indeed, since $\varepsilon_n \rightarrow 0$ uniformly on compact intervals, $1 + \varepsilon_n(v)$ is bounded away from zero for all $v \in [0, 1/a]$ and all sufficiently large n . Moreover, $|f(v)| > 0$, so the logarithm is well-defined, and $\log f(v)$ is bounded on the compact interval $[0, 1/a]$. Hence

$$n(f_n(v) - 1) = \frac{\log f(v)}{1 + \varepsilon_n(v)}$$

is uniformly bounded on $[0, 1/a]$, and therefore so is $n(1 - \Re f_n(v))$. Thus the bounded convergence theorem gives

$$\lim_{n \rightarrow \infty} \int_0^{1/a} n(1 - \Re f_n(v)) dv = \int_0^{1/a} \lim_{n \rightarrow \infty} n(1 - \Re f_n(v)) dv.$$

Therefore

$$\lim_{n \rightarrow \infty} \int_0^{1/a} n(1 - \Re f_n(v)) dv = \int_0^{1/a} -\Re \log f(v) dv.$$

Since the inequality for $\mu_n([-a, a]^c)$ holds for every n , we may take the upper limit on both sides. This gives

$$\limsup_{n \rightarrow \infty} \mu_n([-a, a]^c) \leq \alpha a \int_0^{1/a} -\Re \log f(v) dv.$$

Now, since f is a characteristic function, $|f(v)| \leq 1$. Hence

$$\Re \log f(v) = \log |f(v)| \leq 0,$$

and therefore

$$-\Re \log f(v) = |\Re \log f(v)|.$$

Thus

$$\limsup_{n \rightarrow \infty} \mu_n([-a, a]^c) \leq \alpha a \int_0^{1/a} |\Re \log f(v)| dv.$$

Finally, since $f(0) = 1$, we have $\log f(0) = 0$, and hence

$$\Re \log f(0) = 0.$$

By continuity of $\log f$ near the origin,

$$|\Re \log f(v)| \rightarrow 0 \quad \text{as } v \rightarrow 0.$$

Therefore

$$a \int_0^{1/a} |\Re \log f(v)| dv \leq \sup_{0 \leq v \leq 1/a} |\Re \log f(v)| \rightarrow 0 \quad \text{as } a \rightarrow \infty.$$

Hence

$$\lim_{a \rightarrow \infty} \limsup_{n \rightarrow \infty} \mu_n([-a, a]^c) = 0.$$

Thus the sequence (μ_n) is mass-preserving. □

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Deklaration for anvendelse af generative AI-værktøjer (studerende)
☒ Jeg/vi har benyttet generativ AI som hjælpemiddel/værktøj (sæt kryds) ☐ Jeg/vi har IKKE benyttet generativ AI som hjælpemiddel/værktøj (sæt kryds) <i>Hvis brug af generativ AI er tilladt til eksamen, men du ikke har benyttet det i din opgave, skal du blot krydse af, at du ikke har brugt GAI, og behøver ikke at udfylde resten.</i>
Oplist, hvilke GAI-værktøjer der er benyttet, inkl. link til platformen (hvis muligt): Chatgpt.com Claude.com
Beskriv hvordan generativ AI er anvendt i opgaven: 1) Formål (hvad har du/I brugt værktøjet til) <i>Idegenerering og konceptualisering af beviser og opgaver. Kilde- og litteratursøgning</i> 2) Arbejdsfase (hvornår i arbejdsprocessen har du/I brugt GAI) <i>Under det undersøgende arbejde</i> 3) Hvad gjorde du/I med outputtet (herunder også, om du/I har redigeret outputtet og arbejdet videre med det) <i>Outputtet bestod primært af svar på specifikke spørgsmål og blev brugt som opklarende støtte i arbejdsprocessen.</i> <i>NB. GAI-genereret indhold brugt som kilde i opgaven kræver korrekt brug af citationstegn og kildehenvisning. Læs retningslinjer fra Københavns Universitetsbibliotek på KUnet.</i>